

EC532: Graduate Econometrics 2

Final Exam Study Guide

Multivariate Time Series • Cointegration
Vector Autoregression • VECM
Qualitative & Limited Dependent Variables
Censored, Truncated & Panel Models

Compiled from lecture notes and Problem Sets 3–4

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How to use this guide

This document is a structured study companion to the lecture notes. It is organised so that each topic begins with a *conceptual* explanation — what the idea is and why it matters — before introducing the mathematical machinery. Key derivations are kept compact: the goal is for you to be able to reproduce them, not to admire their length.

A few signposts you will see throughout:

- **Concept boxes** (blue) summarise the core idea before the derivation.
- **Derivation boxes** (gray) walk through key algebraic steps.
- **Intuition boxes** (green) translate symbols into plain language.
- **Problem Set boxes** (orange) flag where a topic appears in PS3 or PS4.
- **Common Pitfall boxes** (red) point out subtle errors.
- **Summary boxes** (purple) give the takeaway at the end of long sections.

The mathematical exposition mirrors the order of the lectures. Notation follows the lecture notes: Y_t, X_t for time-series variables, e_t, u_t, ε_t for error terms, Greek letters for parameters. Subscripts like i for individuals, t for time, j for alternatives.

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1. Dynamic Models: ADL and the Long-Run / Short-Run Decomposition

1.1 Setup: from static to dynamic

A standard cross-section regression $Y_i = \alpha_0 + \alpha_1 X_i + e_i$ implicitly assumes that the entire effect of X on Y is felt instantaneously. In time series this is rarely true. Inflation responds to past unemployment as well as current; consumption depends on past income; output adjusts to past shocks. The natural extension is the **Autoregressive Distributed Lag (ADL)** model:

$$Y_t = \sum_{i=1}^p \rho_i Y_{t-i} + \sum_{j=0}^q \beta_j X_{t-j} + e_t, \quad (1)$$

where e_t is white noise: $\mathbb{E}(e_t) = 0$, $\text{Var}(e_t) = \sigma_e^2$, $\text{Cov}(e_t, e_s) = 0$ for $t \neq s$.

Concept

The ADL model nests two things at once. The autoregressive part ($\rho_i Y_{t-i}$) captures *persistence*: today's Y is partially predictable from its own past. The distributed-lag part ($\beta_j X_{t-j}$) captures *delayed pass-through*: shocks to X feed into Y over multiple periods.

1.2 Motivating example: the Phillips curve

The original Phillips curve is static:

$$\text{infl}_t = \alpha_0 + \alpha_1 \text{unemploy}_t + e_t.$$

But agents form expectations about future inflation, and those expectations themselves depend on the recent history of inflation and unemployment. A more realistic specification might be

$$\text{infl}_t = \alpha_0 + \alpha_1 \text{infl}_{t-1} + \alpha_2 \text{unemploy}_t + \alpha_3 \text{unemploy}_{t-1} + e_t,$$

which is just an ADL(1,1) in $Y = \text{inflation}$ and $X = \text{unemployment}$.

1.3 Estimation and lag selection

If Y_t and X_t are stationary, OLS on (1) is consistent. The two relevant exogeneity conditions are:

- **Strict exogeneity** (for unbiasedness): $\mathbb{E}(e_t | X_1, \dots, X_T) = 0$.
- **Contemporaneous exogeneity** (sufficient for consistency under random sampling, weaker condition): $\mathbb{E}(e_t | X_t) = 0$.

To choose p and q , pick the lag lengths that minimise an information criterion (AIC or BIC). The trade-off is bias (too few lags $\Rightarrow$ omitted dynamics) versus variance (too many lags $\Rightarrow$ noisy estimates).

Common Pitfall

When p and q are large the model has many free parameters, often more than the data can comfortably identify. One practical fix is to impose structure — for example, geometric declining weights $\beta_j = \beta w_j$ with $w_j = (1 - \lambda)\lambda^j$, $0 < \lambda < 1$. This collapses the infinite distributed lag into two parameters, β and λ .

1.4 Short-run vs long-run multipliers

Consider an ADL(2,2):

$$Y_t = \rho_1 Y_{t-1} + \rho_2 Y_{t-2} + \beta_0 X_t + \beta_1 X_{t-1} + \beta_2 X_{t-2} + e_t.$$

Suppose X has been at some baseline forever and X_t rises by one unit *just for one period*. The contemporaneous effect on Y_t is β_0 ; through the autoregressive channel, Y_{t+1} shifts by $\beta_1 + \rho_1 \beta_0$, and so on. This is the **short-run** multiplier: how Y moves over the few periods following a temporary impulse.

The **long-run** multiplier asks the opposite question: if X rises permanently by one unit, by how much does the new equilibrium level of Y exceed the old one?

Long-run multiplier derivation

In the long run all variables sit at constants: $Y_t = Y_{t-1} = Y_{t-2} = Y$ and $X_t = X_{t-1} = X_{t-2} = X$. Take expectations of the ADL(2,2) equation, so $\mathbb{E}(e_t) = 0$:

$$Y = \rho_1 Y + \rho_2 Y + \beta_0 X + \beta_1 X + \beta_2 X.$$

Collect Y on the left:

$$(1 - \rho_1 - \rho_2)Y = (\beta_0 + \beta_1 + \beta_2)X \implies Y = \frac{\beta_0 + \beta_1 + \beta_2}{1 - \rho_1 - \rho_2} X.$$

A unit permanent rise in X shifts the long-run Y by

$$\boxed{\frac{\beta_0 + \beta_1 + \beta_2}{1 - \rho_1 - \rho_2}}.$$

For a general ADL(p, q) the formula is $(\beta_0 + \dots + \beta_q)/(1 - \rho_1 - \dots - \rho_p)$, plus a constant term if one is included.

Problem Set Connection

PS3 Q6. You are given an ARDL(2,2):

$$\hat{Y}_t = 34 - 0.2X_t - 0.075X_{t-1} - 0.025X_{t-2} + 0.43Y_{t-1} + 0.32Y_{t-2}.$$

A 4-unit shock to X at time t raises Y *immediately* by $\beta_0 \cdot 4 = -0.2 \cdot 4 = -0.8$. The change in the new equilibrium level after the system fully adjusts is

$$\frac{\beta_0 + \beta_1 + \beta_2}{1 - \rho_1 - \rho_2} \cdot 4 = \frac{-0.2 - 0.075 - 0.025}{1 - 0.43 - 0.32} \cdot 4 = \frac{-0.3}{0.25} \cdot 4 = -4.8.$$

The negative immediate response gets amplified through the autoregressive lags into a larger long-run decline.

1.5 The Error Correction Model (ECM)

The ADL has two pieces of information mixed together: short-run dynamics (the Δ terms) and the long-run equilibrium relationship between Y and X . A reparametrisation called the **Error Correction Model** cleanly separates them.

ECM derivation from ADL(1,1)

Start from $Y_t = \rho_1 Y_{t-1} + \beta_0 X_t + \beta_1 X_{t-1} + e_t$. Subtract Y_{t-1} from both sides:

$$Y_t - Y_{t-1} = (\rho_1 - 1)Y_{t-1} + \beta_0 X_t + \beta_1 X_{t-1} + e_t.$$

Now add and subtract $\beta_0 X_{t-1}$ on the right:

$$\Delta Y_t = (\rho_1 - 1)Y_{t-1} + \beta_0(X_t - X_{t-1}) + (\beta_0 + \beta_1)X_{t-1} + e_t.$$

Factor $(\rho_1 - 1)$ out of the level terms:

$$\Delta Y_t = \beta_0 \Delta X_t + (\rho_1 - 1) \left[Y_{t-1} + \frac{\beta_0 + \beta_1}{\rho_1 - 1} X_{t-1} \right] + e_t.$$

Define $\phi = -\frac{\beta_0 + \beta_1}{\rho_1 - 1}$ (the long-run coefficient, with positive sign). The ECM form is:

$$\Delta Y_t = \beta_0 \Delta X_t + (\rho_1 - 1) [Y_{t-1} - \phi X_{t-1}] + e_t.$$

Intuition

The bracketed term $v_{t-1} = Y_{t-1} - \phi X_{t-1}$ is the *disequilibrium error* from the previous period — how far Y was from its long-run equilibrium relation with X . The coefficient $(\rho_1 - 1)$ is negative whenever the model is stationary ($|\rho_1| < 1$), so the bracket pulls ΔY_t *back towards* equilibrium. The $\beta_0 \Delta X_t$ term captures pure short-run response. So the ECM literally says: *change in Y = short-run reaction + correction of past disequilibrium*.

Problem Set Connection

PS3 Q2. For the model $y_t = \mu + \rho y_{t-1} + \beta_0 x_t + \beta_1 x_{t-1} + \varepsilon_t$ with long-run equilibrium $y = \alpha + \theta x$:

- Setting $y_t = y_{t-1} = y$, $x_t = x_{t-1} = x$ in the model gives $y(1 - \rho) = \mu + (\beta_0 + \beta_1)x$, so $\theta = (\beta_0 + \beta_1)/(1 - \rho)$ and $\alpha = \mu/(1 - \rho)$.
- To estimate θ directly, run the ECM regression of Δy_t on Δx_t and on $[y_{t-1} - \theta x_{t-1}]$ — but more practically, regress y_t on y_{t-1}, x_t, x_{t-1} and recover θ from the coefficient ratio above.
- The ECM form is exactly the boxed equation above.

1.6 Equivalence of ADL and ECM under stationarity

Because the ECM is just an algebraic rearrangement of the ADL, the following two regressions yield the same fit and equivalent estimates of the same parameters (assuming stationarity, no spurious-regression problem):

- Levels regression:** Y_t on Y_{t-1}, X_t, X_{t-1} .
- ECM regression:** ΔY_t on ΔX_t and $v_{t-1} = Y_{t-1} - \phi X_{t-1}$.

The ECM form has the advantage that the coefficient on v_{t-1} has a direct interpretation as the *speed of adjustment*: a value of -0.3 , say, means 30% of the previous period's disequilibrium is corrected in each period.

1.7 ARMA, ADL, and the Wold representation

The ADL has an observable regressor X . The closely related **ARMA(p,q)** model has only lagged Y and lagged shocks:

$$Y_t = \rho_1 Y_{t-1} + \cdots + \rho_p Y_{t-p} + e_t + \theta_1 e_{t-1} + \cdots + \theta_q e_{t-q}.$$

The MA component $\theta_1 e_{t-1} + \cdots + \theta_q e_{t-q}$ summarises whatever serially-correlated dynamics are not captured by the AR part. Under *invertibility* (the MA polynomial has roots outside the unit circle), an ARMA(p,q) can be rewritten as an AR(∞) — so an ADL with enough lags effectively approximates any ARMA.

Common Pitfall

A key practical asymmetry: when X_t appears as a regressor, you observe it only through period T . To forecast Y_{T+1} you need a forecast of X_{T+1} as well. This is why a pure AR/ARMA model is often easier for short-horizon forecasting than an ADL.

2. Non-Stationary Series, Spurious Regression, and Cointegration

2.1 Why non-stationarity breaks OLS

A series Y_t is **stationary** if its mean, variance, and autocovariances are time-invariant. The simplest non-stationary process is the **random walk**:

$$Y_t = Y_{t-1} + e_t, \quad e_t \text{ white noise.}$$

With $Y_0 = 0$, recursive substitution gives $Y_t = \sum_{s=1}^t e_s$. Variance grows without bound: $\text{Var}(Y_t) = t \cdot \sigma_e^2$. The series wanders, with no tendency to return to a fixed mean.

2.2 The spurious regression problem

Suppose Y_t and X_t are two *independent* random walks — their innovations e_t and e_t^* are completely unrelated. What happens if we naively regress Y_t on X_t ?

Concept

Even though the true relationship is zero, you will typically find a highly significant slope coefficient, an R^2 that looks impressive, and a t -statistic that screams “reject the null of no effect”. This is the **spurious regression** problem (Granger & Newbold, 1974).

To see why, consider $Y_t = \alpha_1 X_t + u_t$ and back out the implied residuals: $u_t = Y_t - \alpha_1 X_t = \sum e_s - \alpha_1 \sum e_s^*$. Two pathologies emerge as the sample grows:

- $\text{Var}(u_t) \rightarrow \infty$. The error variance grows with the sample.
- $\text{Corr}(u_t, u_{t+1}) \rightarrow 1$. The errors are extremely persistent.

OLS minimises the sum of squared errors, but here that sum diverges. Standard error formulas, t -statistics, and F -tests all rely on assumptions that are violated when residuals are non-stationary.

Problem Set Connection

PS3 Q1. The exercise asks you to simulate two independent random walks ten times and regress one on the other. You will find the slope is significant (at conventional 5% levels) far more often than 5% of the time — often above 75%. This is the spurious regression

phenomenon in numerical form.

2.3 Two ways out: differencing and cointegration

There are two responses to non-stationary Y_t and X_t :

1. **Difference everything.** If Y_t is $I(1)$ then $\Delta Y_t = Y_t - Y_{t-1}$ is $I(0)$ (stationary). Run the regression in differences (or in log-differences for variables with multiplicative structure). This works but throws away the long-run information.
2. **Cointegration.** If Y_t and X_t are individually $I(1)$ but a particular linear combination is $I(0)$, that combination is the long-run equilibrium. You can keep the level information by working in levels for the long-run part and in differences for the short-run part. This is the ECM idea applied to non-stationary data.

2.4 What is cointegration?

Concept

Two series are **cointegrated** if each is individually non-stationary, but some linear combination of them is stationary. The drunk-and-the-dog metaphor: each wanders unpredictably, but the leash keeps the distance between them bounded.

The arithmetic rules:

- Linear combination of two $I(1)$ series is generally $I(1)$.
- Linear combination of an $I(1)$ and an $I(0)$ is generally $I(1)$.
- *Exception:* an $I(d)$ and an $I(c)$ series may have a linear combination that is $I(d - c)$. When that combination is stationary, the two series are *cointegrated*.

If Y_t and X_t are cointegrated, we can write

$$z_t = Y_t - \alpha_0 - \alpha_1 X_t \quad \text{is stationary.}$$

The vector $(1, -\alpha_1)$ is the *cointegrating vector*. Real-world examples: nominal exchange rate and the ratio of foreign to domestic price levels (PPP), short-term and long-term interest rates, consumption and income.

2.5 Engle–Granger Representation Theorem

Concept

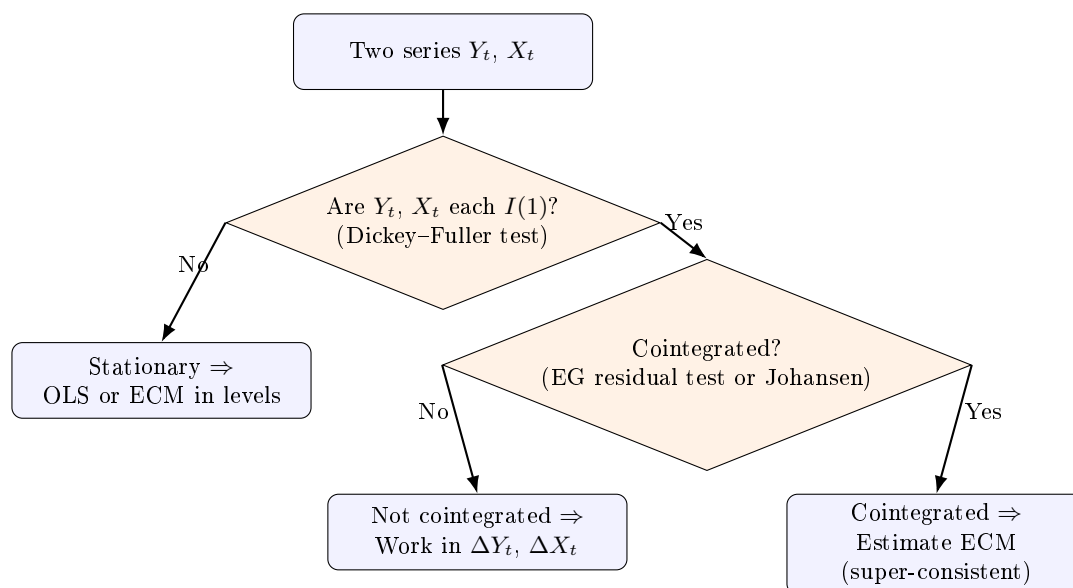
If Y_t and X_t are cointegrated, then the most efficient way to model their joint dynamics is an Error Correction Model. Conversely, if a valid ECM exists, the variables must be cointegrated.

Concretely, the ECM

$$\Delta Y_t = \beta_0 \Delta X_t + (\rho_1 - 1) \left[Y_{t-1} + \frac{\beta_0 + \beta_1}{\rho_1 - 1} X_{t-1} \right] + e_t$$

is internally consistent: ΔY_t is stationary (it's a difference of $I(1)$), ΔX_t is stationary, and the bracket — a linear combination of two $I(1)$ series — is stationary *precisely because Y and X are cointegrated*. Without cointegration the bracket is $I(1)$ and the equation is internally inconsistent.

2.6 Workflow for testing and estimation



2.7 Testing for cointegration: residual-based approach

The idea: if Y_t and X_t are cointegrated, then the residual from a regression of one on the other should be stationary. So:

1. Regress $Y_t = \alpha_0 + \alpha_1 X_t + z_t$ via OLS, save the residuals $\hat{z}_t$.
2. Test whether $\hat{z}_t$ is stationary using a Dickey-Fuller-type test: estimate $\Delta \hat{z}_t = \delta \hat{z}_{t-1} + u_t$ and test $H_0 : \delta = 0$ (i.e. $\hat{z}$ has a unit root, no cointegration) against $H_A : \delta < 0$.

Common Pitfall

The standard Dickey-Fuller critical values do not apply, because $\hat{z}_t$ is constructed from estimated parameters $\hat{\alpha}_0, \hat{\alpha}_1$. The test must use **Engle-Granger critical values**, which are obtained from Monte Carlo simulation and are larger in absolute value than the standard DF critical values.

2.8 Engle-Granger two-stage estimation

Once cointegration is established, the two-stage procedure delivers consistent estimates of the ECM:

1. **Stage 1.** Regress Y_t on X_t in levels:

$$Y_t = \alpha_0 + \alpha_1 X_t + z_t.$$

Save the residuals $\hat{z}_t = Y_t - \hat{\alpha}_0 - \hat{\alpha}_1 X_t$. Verify they are stationary.

2. **Stage 2.** Regress the differenced model with the lagged residual as the disequilibrium term:

$$\Delta Y_t = \beta_0 \Delta X_t + (\rho_1 - 1) \hat{z}_{t-1} + e_t.$$

The coefficient on $\hat{z}_{t-1}$ is the speed of adjustment. *Always check that the residuals from Stage 2 are white noise.*

A useful property: the OLS estimator of α_1 in Stage 1 is **super-consistent**, meaning it converges to the true value at rate T rather than the usual $\sqrt{T}$. The flip side is that standard errors and inference in Stage 1 are non-standard, so t -tests on α_0, α_1 from Stage 1 should not be trusted.

2.9 Engle–Yoo three-step procedure

When the errors e_t in the ECM are themselves serially correlated, the two-step estimator is consistent but inefficient. The Engle–Yoo three-step procedure adds a third regression to refine the cointegrating vector estimates and improve their finite-sample properties. Conceptually it is the same idea: estimate, correct, re-estimate.

Summary

Quick decision rule:

- Stationary Y_t, X_t : estimate ADL or ECM via OLS in levels.
- Non-stationary, no cointegration: regress ΔY_t on ΔX_t (levels regression is spurious).
- Non-stationary, cointegrated: ECM via Engle–Granger 2-stage (or Johansen for multivariate systems).

3. Granger Causality

3.1 The concept

Concept

X is said to **Granger-cause** Y if past values of X help predict Y *beyond* what past values of Y alone can predict. The name is a slight misnomer: this is a statement about predictive content, not about true causation in the philosophical sense.

The intuition: even if X and Y are simultaneously determined, the past of X might contain information about future Y that the past of Y does not contain.

3.2 The test

The standard implementation regresses Y on its own lags and on lags of X :

$$Y_t = \alpha + \rho_1 Y_{t-1} + \cdots + \rho_p Y_{t-p} + \beta_1 X_{t-1} + \cdots + \beta_q X_{t-q} + e_t.$$

Often $p = q$ in practice. The null hypothesis “ X does not Granger-cause Y ” is

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_q = 0,$$

and we test it with a standard F -test on the joint significance of the lags of X . Failing to reject H_0 is evidence of no Granger causality.

Common Pitfall

The test assumes *stationarity*. With non-stationary or cointegrated series, the standard F -distribution may not be valid; one then works in differences, or uses Granger causality tests within a VECM framework.

3.3 Stock prices and dividends under EMH

A textbook illustration: under the Efficient Market Hypothesis, the current stock price embeds expectations of future dividends:

$$(1+r)P_t = \mathbb{E}[P_{t+1} + D_{t+1}] \implies P_t = \frac{\mathbb{E}(P_{t+1} + D_{t+1})}{1+r}.$$

Today’s price reflects all information about future dividends. Therefore:

- P_t *should* Granger-cause D_{t+1} : today's price contains information about future dividends.
- D_{t+1} *should not* Granger-cause P_{t+1} : news about dividends one period ahead is already in P_t , so D_{t+1} adds nothing.

A real-world application from the lecture: oil price changes Granger-cause recessions; the reverse direction is much weaker.

3.4 An applied example: tourism and terrorism

A study of transnational terrorism and tourism flows tested both directions:

- $Terrorism \rightarrow Tourism$: significant. Attacks reduce arrivals.
- $Tourism \rightarrow Terrorism$: not significant.

This kind of asymmetric finding is exactly what Granger causality is designed to uncover.

Problem Set Connection

PS3 Q4(a). Given the VAR estimates, examine whether each variable's lag is significant in the *other* variable's equation. The coefficient on $Y_{(t-1),2}$ in the equation for $\hat{Y}_{t,1}$ is 0.440 with standard error 0.10, giving a t -statistic of 4.4 — highly significant. So Y_2 Granger-causes Y_1 . The coefficient on $Y_{(t-1),1}$ in the $\hat{Y}_{t,2}$ equation is 0.241 with standard error 0.12, giving $t \approx 2.0$ — borderline significant. The system has stronger feedback in one direction.

4. Vector Autoregression (VAR)

4.1 Why we need VARs

In a single-equation regression of Y_t on X_t , an implicit assumption is that X_t is exogenous — shocks to Y do not feed back into X . In macroeconomics this is rarely true: GDP and interest rates respond to each other; consumption and income are jointly determined.

Concept

A **Vector Autoregression (VAR)** treats a set of variables symmetrically, regressing each on lags of all variables (including its own). The result is a system of equations that lets dynamics flow in all directions.

4.2 Structural form vs reduced form

Structural form (with contemporaneous interactions):

$$\begin{aligned} Y_{1t} &= \beta_{10} + \beta_{11}Y_{1,t-1} + \beta_{12}Y_{2,t-1} + \beta_{13}Y_{2t} + u_{1t}, \\ Y_{2t} &= \beta_{20} + \beta_{21}Y_{1,t-1} + \beta_{22}Y_{2,t-1} + \beta_{23}Y_{1t} + u_{2t}. \end{aligned}$$

The contemporaneous terms $\beta_{13}Y_{2t}$ and $\beta_{23}Y_{1t}$ create a simultaneity problem: Y_{1t} and u_{1t} are no longer independent of Y_{2t} , so OLS is biased and the parameters are **not identified** without additional restrictions.

Reduced form (no contemporaneous terms):

$$\begin{aligned} Y_{1t} &= \alpha_{10} + \alpha_{11}Y_{1,t-1} + \alpha_{12}Y_{2,t-1} + e_{1t}, \\ Y_{2t} &= \alpha_{20} + \alpha_{21}Y_{1,t-1} + \alpha_{22}Y_{2,t-1} + e_{2t}. \end{aligned}$$

Each equation has only lagged variables on the right-hand side, so OLS is consistent equation by equation. The reduced form is identified and estimable.

From structural to reduced form

Substitute the second structural equation into the first to eliminate Y_{2t} :

$$Y_{1t} = \beta_{10} + \beta_{11}Y_{1,t-1} + \beta_{12}Y_{2,t-1} + \beta_{13}[\beta_{20} + \beta_{21}Y_{1,t-1} + \beta_{22}Y_{2,t-1} + \beta_{23}Y_{1t} + u_{2t}] + u_{1t}.$$

Collect Y_{1t} :

$$Y_{1t}(1 - \beta_{13}\beta_{23}) = (\beta_{10} + \beta_{13}\beta_{20}) + Y_{1,t-1}(\beta_{11} + \beta_{13}\beta_{21}) + Y_{2,t-1}(\beta_{12} + \beta_{13}\beta_{22}) + u_{1t} + \beta_{13}u_{2t}.$$

Divide through by $(1 - \beta_{13}\beta_{23})$ to get the reduced-form coefficients $\alpha_{10}, a_{11}, a_{12}$ and reduced-form error e_{1t} . Note that the reduced-form errors are linear combinations of the structural errors, so they are typically *correlated* across equations.

4.3 Compact matrix notation

Stacking the variables $Y_t = (Y_{1t}, Y_{2t})'$, the structural and reduced forms become:

$$BY_t = \beta_0 + \beta_1 Y_{t-1} + u_t \quad (\text{structural}), \quad Y_t = A_0 + A_1 Y_{t-1} + e_t \quad (\text{reduced}),$$

with $A_0 = B^{-1}\beta_0$, $A_1 = B^{-1}\beta_1$, $e_t = B^{-1}u_t$. The reduced form is what we actually estimate; the structural form is what we want to interpret.

Problem Set Connection

PS3 Q5. For $p = 2$ variables and $k = 3$ lags, a VAR(3) is:

$$\begin{aligned} Y_{1t} &= \alpha_{10} + a_{11}^{(1)}Y_{1,t-1} + a_{12}^{(1)}Y_{2,t-1} + a_{11}^{(2)}Y_{1,t-2} + a_{12}^{(2)}Y_{2,t-2} + a_{11}^{(3)}Y_{1,t-3} + a_{12}^{(3)}Y_{2,t-3} + e_{1t}, \\ Y_{2t} &= \alpha_{20} + a_{21}^{(1)}Y_{1,t-1} + a_{22}^{(1)}Y_{2,t-1} + a_{21}^{(2)}Y_{1,t-2} + a_{22}^{(2)}Y_{2,t-2} + a_{21}^{(3)}Y_{1,t-3} + a_{22}^{(3)}Y_{2,t-3} + e_{2t}. \end{aligned}$$

The superscript (ℓ) on $a_{ij}^{(\ell)}$ indexes the lag, while subscripts i, j index the equation and the regressor.

4.4 Stationarity in a VAR via eigenvalues

For a VAR(1) $Y_t = m + AY_{t-1} + e_t$, stationarity depends on the matrix A . The trick is to diagonalise.

Diagonalisation gives independent AR(1) processes

Let $A = C\Lambda C^{-1}$ where $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$ holds the eigenvalues and C holds the eigenvectors. Define the transformed series $z_t = C^{-1}Y_t$. Pre-multiply the VAR by C^{-1} :

$$C^{-1}Y_t = C^{-1}m + C^{-1}AY_{t-1} + C^{-1}e_t.$$

Insert $A = C\Lambda C^{-1}$, so $C^{-1}A = \Lambda C^{-1}$:

$$z_t = m^* + \Lambda z_{t-1} + e_t^*,$$

with $m^* = C^{-1}m$, $e_t^* = C^{-1}e_t$. Since Λ is diagonal, this is n independent AR(1) equations:

$$z_{it} = m_i^* + \lambda_i z_{i,t-1} + e_{it}^*, \quad i = 1, \dots, n.$$

Each is stationary iff $|\lambda_i| < 1$.

Because $Y_t = Cz_t$ is a linear combination of the z 's, the stationarity properties of Y_t are determined by the z 's.

4.5 Three cases (two-variable system)

1. **Both** $|\lambda_1|, |\lambda_2| < 1$. Both z 's stationary; both Y 's stationary. Use OLS or VECM.
2. $\lambda_1 = 1, |\lambda_2| < 1$. z_1 is $I(1)$ but z_2 is $I(0)$. The Y 's are non-stationary (since each is a linear combination of z_1 and z_2 , and any combination involving z_1 inherits its non-stationarity). *However*, the particular combination $z_2 = (\text{certain rows of } C^{-1}) \cdot Y_t$ is *stationary*. So Y_1 and Y_2 are **cointegrated**. There is one cointegrating relation.
3. $\lambda_1 = \lambda_2 = 1$. Both z 's non-stationary, both Y 's non-stationary, no stationary linear combination exists $\Rightarrow$ no cointegration. Work in ΔY_t .

4.6 The $(I - A)$ matrix and rank

A clean way to read off the cases above: examine $(I - A)$. Its eigenvalues are $(1 - \lambda_i)$, and it shares eigenvectors with A .

- $\text{rank}(I - A) = 0$: all eigenvalues of A equal 1 $\Rightarrow$ all variables non-stationary, no cointegration.
- $\text{rank}(I - A) = n$ (full rank): all eigenvalues differ from 1 $\Rightarrow$ all variables stationary.
- $0 < \text{rank}(I - A) = s < n$: s eigenvalues differ from 1, the rest equal 1 $\Rightarrow$ variables non-stationary but s cointegrating relations exist.

Mnemonic: the rank of $(I - A)$ equals the number of cointegrating vectors.

4.7 Vector Error Correction Model (VECM)

The single-equation ECM extends to the multivariate case. For a VAR(1):

VECM from VAR(1)

Subtract Y_{t-1} from both sides of $Y_t = m + AY_{t-1} + e_t$:

$$\Delta Y_t = m + (A - I)Y_{t-1} + e_t = m - (I - A)Y_{t-1} + e_t.$$

Define $\Pi = I - A$. The VECM is

$$\Delta Y_t = m - \Pi Y_{t-1} + e_t.$$

The rank of Π tells us how many cointegrating relations there are.

For a VAR(2) the algebra adds one extra step:

VECM from VAR(2)

Start from $Y_t = m + A_1 Y_{t-1} + A_2 Y_{t-2} + e_t$. Subtract Y_{t-1} :

$$\Delta Y_t = m + (A_1 - I)Y_{t-1} + A_2 Y_{t-2} + e_t.$$

Add and subtract $(A_1 - I)Y_{t-2}$ on the right:

$$\Delta Y_t = m + (A_1 - I)(Y_{t-1} - Y_{t-2}) + (A_1 - I)Y_{t-2} + A_2 Y_{t-2} + e_t.$$

Combine the Y_{t-2} coefficients:

$$\Delta Y_t = m + (A_1 - I)\Delta Y_{t-1} + (A_1 + A_2 - I)Y_{t-2} + e_t.$$

Define $\Pi = I - A_1 - A_2$ (sign flipped for convention) and $B_1 = A_1 - I$. Then

$$\Delta Y_t = m + B_1 \Delta Y_{t-1} - \Pi Y_{t-2} + e_t.$$

The general VAR(p) gives

$$\Delta Y_t = m + B_1 \Delta Y_{t-1} + \cdots + B_{p-1} \Delta Y_{t-p+1} - \Pi Y_{t-p} + e_t,$$

with $\Pi = I - A_1 - A_2 - \cdots - A_p$.

4.8 The $\Pi = \alpha\beta'$ decomposition

When $\text{rank}(\Pi) = s$ with $0 < s < k$, Π admits a decomposition that separates the long-run information from the speed-of-adjustment information:

$$\Pi_{k \times k} = \alpha_{k \times s} \cdot \beta'_{s \times k}.$$

Intuition

- β' holds the **cointegrating vectors**. Each row of β' is one cointegrating relation: $\beta'_j Y_{t-1}$ is a stationary disequilibrium term.
- α holds the **adjustment loadings**. Each row of α tells us, for one variable in the system, how strongly it responds to each disequilibrium.

A 3-variable system with rank 2 has α of size 3×2 and β' of size 2×3 — two cointegrating relations, three speeds of adjustment per relation.

Identification. The decomposition $\Pi = \alpha\beta'$ is not unique: for any invertible $s \times s$ matrix G , we can write $\Pi = (\alpha G)(G^{-1}\beta')$. To pin down the cointegrating vectors uniquely, a standard normalisation sets the leading $s \times s$ block of β' to the identity (i.e. each cointegrating relation is solved for one “representative” variable).

4.9 The Johansen test

The Johansen procedure is a likelihood-based test for the rank of Π — equivalently, the number of cointegrating relations.

Concept

Logic. Estimate $\hat{\Pi}$ via maximum likelihood (assuming normal errors), compute its eigenvalues, and test sequentially: how many of these eigenvalues are non-zero?

The estimated eigenvalues $\hat{\lambda}_1 \geq \hat{\lambda}_2 \geq \cdots \geq \hat{\lambda}_k$ are unlikely to be exactly zero in finite samples even when their true values are. The Johansen test statistics are:

$$\text{Trace test:} \quad \text{LR}_{\text{trace}}(r) = -T \sum_{i=r+1}^k \ln(1 - \hat{\lambda}_i),$$

$$\text{Maximum eigenvalue test:} \quad \text{LR}_{\text{max}}(r) = -T \ln(1 - \hat{\lambda}_{r+1}).$$

The trace test has H_0 : rank = r against H_A : rank $> r$. The max test has H_A : rank = $r + 1$. Both use *non-standard* critical values that depend on the deterministic terms (constant, trend) included.

Sequential testing procedure:

1. Test $H_0 : r = 0$ against $H_A : r > 0$ (or $r = 1$). If you fail to reject, conclude there is no cointegration.
2. If you reject, test $H_0 : r = 1$ against $H_A : r > 1$. If you fail to reject, there is exactly one cointegrating relation.
3. Continue until you fail to reject.

Problem Set Connection

PS3 Q3. You have a VAR with four variables and the following Johansen output:

r	$\lambda_{\max}$	critical value
0	43.52	31.18
1	21.98	25.19
2	15.04	20.28
3	8.81	14.06

(a) With four variables, the maximum number of cointegrating relations is $k - 1 = 3$. (Rank k would mean all variables are stationary, contradicting the assumption that all are non-stationary.)

(b) Sequential test:

- $r = 0$: $43.52 > 31.18 \Rightarrow$ reject. There is at least one cointegrating relation.
- $r = 1$: $21.98 < 25.19 \Rightarrow$ fail to reject. Stop.

Conclusion: **exactly one cointegrating relation.**

4.10 Estimator choice summary

Summary

For a k -variable VAR system:

- All variables stationary: estimate the VAR by OLS equation by equation.
- All non-stationary, no cointegration: estimate VAR in differences (ΔY_t).
- All non-stationary, s cointegrating relations ($0 < s < k$): estimate the VECM via Johansen's MLE.

The lecturer's note "giving a meaning to the estimates is challenging" refers especially to the third case: the cointegrating vectors β are identified up to an orthogonal rotation, so attaching economic meaning to specific entries requires a normalisation choice that is partly arbitrary.

5. Impulse Response, Variance Decomposition, and Forecasting

5.1 The point of these tools

Concept

Once you have estimated a VAR (or VECM), you have a system of equations that captures the dynamic interactions among variables. But coefficients alone are hard to interpret economically. Three tools turn the estimated system into something you can read:

- **Impulse Response Functions (IRFs)** trace how a one-time shock propagates through the system over time.
- **Forecast Error Variance Decomposition (FEVD)** apportions forecast uncertainty across the system's shocks.
- **Forecasting** produces point predictions and (via bootstrap) confidence bands for future values.

5.2 Impulse Response Function: definition and example

Consider the VAR(1) $Y_t = A_0 + A_1 Y_{t-1} + e_t$. By repeated substitution, Y_t can be written as a moving average of past shocks:

$$Y_t = m + e_t + A_1 e_{t-1} + A_1^2 e_{t-2} + A_1^3 e_{t-3} + \dots$$

The IRF at horizon h is the matrix A_1^h : its (i, j) entry tells us the response of Y_i at time $t + h$ to a one-unit shock to e_j at time t .

A worked numerical example

Let

$$A_1 = \begin{pmatrix} 0.4 & 0.1 \\ 0.2 & 0.5 \end{pmatrix}, \quad \Sigma_e = \begin{pmatrix} 16 & 14 \\ 14 & 25 \end{pmatrix}, \quad Y_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

A one standard deviation shock to e_1 is $\sqrt{16} = 4$. *Provisionally* assuming $e_2 = 0$ and all subsequent shocks are zero:

$$\begin{aligned} Y_1 &= e_1 = (4, 0)', \\ Y_2 &= A_1 Y_1 = (0.4 \cdot 4 + 0.1 \cdot 0, 0.2 \cdot 4 + 0.5 \cdot 0)' = (1.6, 0.8)', \\ Y_3 &= A_1 Y_2 = (0.4 \cdot 1.6 + 0.1 \cdot 0.8, 0.2 \cdot 1.6 + 0.5 \cdot 0.8)' = (0.72, 0.72)', \\ Y_4 &\approx (0.36, 0.504)', \\ Y_5 &\approx (0.194, 0.324)'. \end{aligned}$$

The shock decays geometrically. Both variables eventually return to zero because A_1 's eigenvalues are inside the unit circle.

5.3 The orthogonalisation problem

The naive IRF above has a serious issue: $\text{Cov}(e_1, e_2) = 14 \neq 0$. Setting $e_2 = 0$ while e_1 jumps is unrealistic, because the two shocks tend to move together. We need *independent* shocks before we can sensibly trace a shock to one without affecting the others.

Cholesky orthogonalisation

Construct two new shocks u_1, u_2 with $\mathbb{E}(u_i) = 0$, $\text{Var}(u_i) = 1$, $\text{Cov}(u_1, u_2) = 0$, and re-express e_1, e_2 in terms of them.

Step 1: standardise e_1 to make u_1 .

$$u_{1t} = \frac{e_{1t}}{\text{s.d.}(e_{1t})} = \frac{e_{1t}}{4}, \quad \text{Var}(u_{1t}) = \frac{16}{16} = 1.$$

Step 2: residualise e_2 against e_1 and standardise. The regression coefficient is $b_{21} = \text{Cov}(e_1, e_2) / \text{Var}(e_1) = 14/16 = 0.875$. The residual $e_{2t} - b_{21}e_{1t}$ has variance

$$\text{Var}(e_{2t})(1 - R^2) = 25 \cdot \left(1 - \frac{14^2}{16 \cdot 25}\right) = 25 \cdot 0.51 = 12.75,$$

so its standard deviation $S_{21} = \sqrt{12.75} \approx 3.57$. Define

$$u_{2t} = \frac{e_{2t} - b_{21}e_{1t}}{S_{21}}.$$

By construction u_{1t} and u_{2t} are uncorrelated and each has variance one.

Step 3: invert.

$$e_{1t} = 4u_{1t}, \quad e_{2t} = S_{21}u_{2t} + b_{21} \cdot \frac{e_{1t}}{4} = 3.57u_{2t} + 3.5u_{1t}.$$

Now a shock to u_{1t} moves both e_1 and e_2 in a self-consistent way; a shock to u_{2t} moves only e_2 .

In matrix form: write $\Sigma_e = PP'$ with P lower triangular (the **Cholesky decomposition**) and let $u_t = P^{-1}e_t$. Then $\text{Cov}(u_t) = P^{-1}\Sigma_e(P^{-1})' = I$.

Common Pitfall

The order of variables matters. Putting Y_1 first in the system and applying Cholesky imposes that u_{1t} shocks affect both variables contemporaneously, while u_{2t} shocks affect only Y_2 contemporaneously. This is a substantive economic assumption — you are saying Y_1 is “most exogenous” of the two. Different orderings give different IRFs. Robustness to ordering is a standard sensitivity check.

5.4 Forecasting from a VAR

Given the estimated VAR $Y_t = A_1Y_{t-1} + \dots + A_pY_{t-p} + e_t$, the h -step-ahead forecast is built recursively:

- For $h = 1$: $\hat{Y}_{T+1} = \hat{A}_1Y_T + \dots + \hat{A}_pY_{T-p+1}$, setting $e_{T+1} = 0$.
- For $h > 1$: replace any Y_{T+j} with $j > 0$ by its forecast $\hat{Y}_{T+j}$. The error contributions $e_{T+1}, \dots, e_{T+h}$ are all set to their unconditional mean of zero.

Problem Set Connection

PS3 Q4(b). Given the VAR estimates and the data through $t = 10$, the one-step-ahead forecast for period 11 uses observed Y_{10} :

$$\begin{aligned} \hat{Y}_{11,1} &= -7.038 + 0.019 \cdot Y_{10,1} + 0.440 \cdot Y_{10,2} = -7.038 + 0.019 \cdot 10 + 0.440 \cdot 20 \\ &= -7.038 + 0.19 + 8.80 = 1.952. \end{aligned}$$

$$\hat{Y}_{11,2} = -7.657 + 0.241 \cdot Y_{10,1} + 0.283 \cdot Y_{10,2} = -7.657 + 2.41 + 5.66 = 0.413.$$

For period 12, plug the period-11 *forecasts* (not actual values) back into the equation:

$$\begin{aligned}\hat{Y}_{12,1} &= -7.038 + 0.019 \cdot 1.952 + 0.440 \cdot 0.413 \approx -6.819. \\ \hat{Y}_{12,2} &= -7.657 + 0.241 \cdot 1.952 + 0.283 \cdot 0.413 \approx -7.069.\end{aligned}$$

5.5 Forecast error and its variance

Write the VAR in MA(∞) form: $Y_t = \mu + \sum_{j=0}^{\infty} \Phi_j e_{t-j}$ with $\Phi_0 = I$. Then

$$Y_{T+h} - \hat{Y}_{T+h} = \sum_{j=0}^{h-1} \Phi_j e_{T+h-j} + \underbrace{\text{terms from } \Phi_{h+i} - \hat{\Phi}_{h+i}}_{\text{disregarded in most applications}}.$$

The dominant term is the sum of unforecastable future shocks weighted by the MA coefficients. Its variance is

$$\text{Var}(Y_{T+h} - \hat{Y}_{T+h}) = \sum_{j=0}^{h-1} \Phi_j \Sigma_e \Phi_j'.$$

Intuition

The forecast error variance for, say, Y_1 depends not only on the variance of Y_1 's own equation error but also on errors in the *other* equations of the system. Information about Y_2 's shock matters for predicting Y_1 because of dynamic feedback.

5.6 Forecast Error Variance Decomposition

After orthogonalising shocks via Cholesky ($u_t = P^{-1}e_t$, $\text{Cov}(u_t) = I$), rewrite the MA representation in terms of u_t :

$$Y_t = \mu + \sum_{j=0}^{\infty} \Psi_j u_{t-j}, \quad \Psi_j = \Phi_j P.$$

The forecast error variance for variable i at horizon h becomes

$$\text{Var}(Y_{i,T+h} - \hat{Y}_{i,T+h}) = \sum_{j=0}^{h-1} \sum_{\ell=1}^k \psi_{j,i\ell}^2.$$

The share of this variance attributable to shocks in the r -th equation is

$$\text{Share}_{i \leftarrow r}(h) = \frac{\sum_{j=0}^{h-1} \psi_{j,ir}^2}{\sum_{j=0}^{h-1} \sum_{\ell=1}^k \psi_{j,i\ell}^2}.$$

This shares-of-variance breakdown by source is the **Forecast Error Variance Decomposition**. At very short horizons, most of Y_i 's forecast variance comes from its own shock; at longer horizons, shocks from other equations contribute more. A typical FEVD output is a table or stacked-bar plot showing how each variable's prediction uncertainty is allocated across $u_1, u_2, \dots, u_k$ at horizons $h = 1, 2, 4, 8, \dots$

5.7 Bootstrap confidence intervals for forecasts

Analytical confidence bands for VAR forecasts are awkward because they need to account for both shock uncertainty and parameter estimation uncertainty. The bootstrap is a clean alternative.

Procedure:

1. Estimate the VAR; obtain $\hat{A}$ and residuals $\hat{e}_t$.
2. Generate a bootstrap sample by drawing residuals $\hat{e}_t^{(b)}$ randomly with replacement, and constructing $Y_t^{(b)}$ recursively from the estimated $\hat{A}$ and the bootstrap residuals.
3. Re-estimate the VAR on each bootstrap sample, generate forecasts.
4. Repeat (say) 500 times. The 5th and 95th percentiles of the bootstrap forecast distribution form a 90% confidence interval.

5.8 Applications mentioned in lecture

- **Tourism and terrorism:** Granger causality from terrorism to tourism is significant; reverse direction is not.
- **PPP via VECM:** testing the long-run relationship between exchange rates and price ratios.
- **International bond yields:** cointegration tested via Johansen, and FEVD used to decompose forecast variance across countries.

6. Qualitative Dependent Variables: LPM, Probit, Logit

6.1 When the dependent variable is categorical

Concept

A **qualitative** variable takes values that label categories rather than measure quantities. The numbers attached to categories may carry no quantitative meaning at all. A continuous regression of Y on X no longer makes sense, because changes in Y are constrained to a discrete set.

Three flavours appear in practice:

- **Binary:** two categories, encoded 0/1. Examples: labour force participation, mortgage default, heating choice (gas/coal). → probit, logit.
- **Ordered:** multiple ordered categories. Example: education 0 = middle school, 1 = high school, 2 = undergrad+. Numbers carry ordering information. → ordered probit/logit.
- **Unordered (nominal):** multiple categories with no natural ranking. Example: occupation 0 = engineer, 1 = lawyer, 2 = economist. The numbers are arbitrary labels. → multinomial logit.

A **limited** dependent variable is technically continuous but observed only over part of its range (e.g. income topcoded at \$250k). This is censoring and is treated separately in Part 9.

6.2 The Linear Probability Model

The simplest approach is just OLS:

$$Y_i = \alpha_0 + \alpha_1 X_i + u_i, \quad Y_i \in \{0, 1\}.$$

Because $\mathbb{E}(Y_i) = 1 \cdot \text{P}(Y_i = 1) + 0 \cdot \text{P}(Y_i = 0) = \text{P}(Y_i = 1)$, the regression equation has a probabilistic reading:

$$\text{P}(Y_i = 1 \mid X_i) = \alpha_0 + \alpha_1 X_i.$$

The slope α_1 is the change in the probability of $Y = 1$ per unit change in X . This is the **Linear Probability Model (LPM)**.

6.2.1 Properties of the error term

Error structure of the LPM

The error is $u_i = Y_i - \mathbb{E}(Y_i) = Y_i - (\alpha_0 + \alpha_1 X_i)$. It takes only two values:

$$u_i = \begin{cases} 1 - \alpha_0 - \alpha_1 X_i & \text{with probability } \alpha_0 + \alpha_1 X_i, \\ -\alpha_0 - \alpha_1 X_i & \text{with probability } 1 - \alpha_0 - \alpha_1 X_i. \end{cases}$$

Then $\mathbb{E}(u_i) = 0$ as required, and

$$\begin{aligned} \text{Var}(u_i) &= \mathbb{E}(u_i^2) \\ &= (1 - \alpha_0 - \alpha_1 X_i)^2(\alpha_0 + \alpha_1 X_i) + (-\alpha_0 - \alpha_1 X_i)^2(1 - \alpha_0 - \alpha_1 X_i) \\ &= (1 - \alpha_0 - \alpha_1 X_i)(\alpha_0 + \alpha_1 X_i)[(1 - \alpha_0 - \alpha_1 X_i) + (\alpha_0 + \alpha_1 X_i)] \\ &= (1 - \alpha_0 - \alpha_1 X_i)(\alpha_0 + \alpha_1 X_i) = p_i(1 - p_i). \end{aligned}$$

The variance *depends on* X_i — the LPM is intrinsically heteroskedastic. Use Feasible GLS (or robust standard errors).

6.2.2 Why the LPM is unsatisfactory

Common Pitfall

Two problems make the LPM unappealing despite its simplicity:

1. **Predictions outside** $[0, 1]$. The fitted line $\hat{\alpha}_0 + \hat{\alpha}_1 X_i$ is not bounded; for sufficiently large or small X , the predicted “probability” is negative or exceeds one.
2. **Constant marginal effect**. Whether $X_i = 0$ or $X_i = 100$, the LPM says the partial effect on $P(Y = 1)$ is the same. This contradicts intuition: someone whose probability is already 0.99 cannot have it raised much further by an additional unit of X .

A textbook example: estimating heating choice (1 = gas, 0 = coal) on log relative price gives $\hat{\alpha}_0 = 0.784$, $\hat{\alpha}_1 = -0.243$. At $X = 4$, the predicted probability is $0.784 - 4 \cdot 0.243 = -0.188$ — meaningless.

6.3 Probit and logit: bounded link functions

The fix is to wrap the linear index $X_i' \beta$ inside a function $F(\cdot)$ that maps $\mathbb{R}$ to $[0, 1]$:

$$P(Y_i = 1 \mid X_i) = F(X_i' \beta).$$

Two standard choices:

- **Probit**: $F(z) = \Phi(z)$, the standard normal CDF.

$$\Phi(z) = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt.$$

The density is $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$.

- **Logit**: $F(z) = \frac{e^z}{1 + e^z} = \frac{1}{1 + e^{-z}}$, the logistic CDF. The density satisfies the elegant identity $f(z) = F(z)(1 - F(z))$, which simplifies the FOC.

The two are nearly indistinguishable in fit, with logit having slightly heavier tails. Coefficient magnitudes differ (logit coefficients are roughly $\pi/\sqrt{3} \approx 1.81$ times the corresponding probit coefficients), but estimated probabilities and marginal effects line up closely.

6.4 Maximum likelihood estimation

Since the LPM's OLS approach doesn't generalise to the bounded F , we estimate β by maximum likelihood. With $Y_i \in \{0, 1\}$ and $F_i = F(X_i'\beta)$:

$$P(Y_i = y_i) = F_i^{y_i}(1 - F_i)^{1-y_i}.$$

Assuming i.i.d. observations, the likelihood is

$$\mathcal{L}(\beta) = \prod_{i=1}^n F_i^{y_i}(1 - F_i)^{1-y_i},$$

and the log-likelihood is

$$\ln \mathcal{L}(\beta) = \sum_{i=1}^n [y_i \ln F_i + (1 - y_i) \ln(1 - F_i)].$$

First-order condition

Writing $f_i = F_i'$ for the density, the score is

$$\frac{\partial \ln \mathcal{L}}{\partial \beta} = \sum_i \left[\frac{y_i}{F_i} f_i X_i - \frac{1 - y_i}{1 - F_i} f_i X_i \right] = \sum_i \frac{y_i - F_i}{F_i(1 - F_i)} f_i X_i.$$

Setting this to zero gives the MLE.

For **probit** ($f_i = \phi_i$), the FOC has no closed form $\Rightarrow$ numerical optimisation required.

For **logit** ($f_i = F_i(1 - F_i)$), the $F_i(1 - F_i)$ factors cancel and the FOC simplifies enormously:

$$\sum_i (y_i - F_i) X_i = 0.$$

This is just an orthogonality condition between the residual $y_i - F_i$ and the regressors X_i — conceptually similar to OLS, except F_i is non-linear in β . Newton-Raphson converges quickly because the log-likelihood is globally concave for logit (and probit).

Problem Set Connection

PS4 Q1. Suppose $u_i \sim \text{Uniform}[-1, 1]$ and $|x_i'\beta| < 1$. Then

$$P(\delta_i = 1 \mid x_i) = P(x_i'\beta + u_i > 0) = P(u_i > -x_i'\beta) = \frac{1 - (-x_i'\beta)}{2} = \frac{1 + x_i'\beta}{2}.$$

The likelihood is

$$\mathcal{L} = \prod_i \left(\frac{1 + x_i'\beta}{2} \right)^{\delta_i} \left(\frac{1 - x_i'\beta}{2} \right)^{1-\delta_i}.$$

This is the same general structure as probit/logit but with the uniform CDF as the link.

6.5 Newton–Raphson method

Concept

Numerical maximisation of $\ln \mathcal{L}$ is like climbing a hill: at each step, take the local quadratic approximation to the objective and jump to its maximum. The recipe needs the gradient (score) and the curvature (Hessian or its expected value, the information matrix).

Define the score and information matrix:

$$S(\beta) = \frac{\partial \ln \mathcal{L}}{\partial \beta}, \quad R(\beta) = -\mathbb{E} \left(\frac{\partial^2 \ln \mathcal{L}}{\partial \beta \partial \beta'} \right).$$

The update rule is

$$\beta^{(k+1)} = \beta^{(k)} + [R(\beta^{(k)})]^{-1} S(\beta^{(k)}).$$

Iterate from an initial guess (commonly the OLS estimate) until $\beta^{(k+1)} \approx \beta^{(k)}$.

6.6 Latent variable interpretation

Discrete choice models often have a clean interpretation through an unobserved continuous variable.

Concept

Female labour force participation example. Each woman has a potential market wage $W_i^* = X_i' \beta + u_i$ and a reservation wage r . She participates iff $W_i^* \geq r$.

The observable outcome is

$$Y_i = \begin{cases} 1 & \text{if } u_i \geq r - X_i' \beta, \\ 0 & \text{if } u_i < r - X_i' \beta. \end{cases}$$

Then

$$P(Y_i = 1) = P(u_i \geq r - X_i' \beta) = 1 - F_u(r - X_i' \beta).$$

If $u_i \sim N(0, 1)$: $P(Y_i = 1) = 1 - \Phi(r - X_i' \beta) = \Phi(X_i' \beta - r)$ — this is probit, with the constant absorbing the threshold r .

If u_i has a logistic distribution: probabilities take the logit form.

Intuition

The two distributional choices look almost identical except in the tails: the logistic has fatter tails, so logit predicts slightly larger response probabilities at extreme X values. With normal data, probit is preferred. With long-tailed underlying processes, logit may fit better.

6.7 Interpreting the coefficients: marginal effects**Common Pitfall**

The MLE coefficient $\hat{\beta}_j$ does *not* answer the question “how much does $P(Y = 1)$ change when X_j rises by one unit?”. It only tells you the sign of the change. For magnitude, you must compute a marginal effect.

By the chain rule:

$$\frac{\partial P(Y_i = 1)}{\partial X_{ji}} = \frac{\partial F(X_i' \beta)}{\partial (X_i' \beta)} \cdot \frac{\partial X_i' \beta}{\partial X_{ji}} = f(X_i' \beta) \cdot \beta_j.$$

Since $f > 0$ everywhere, $\text{sgn}(\text{marginal effect}) = \text{sgn}(\beta_j)$. The magnitude depends on *where* you evaluate X_i . Three standard choices:

1. **Marginal effect at the mean:** evaluate at $\bar{X} = \frac{1}{n} \sum_i X_i$, giving $f(\bar{X}'\hat{\beta}) \cdot \hat{\beta}_j$. Easy to compute, but does not represent any particular individual.
2. **Average marginal effect:** compute the marginal effect for each individual and average: $\frac{1}{n} \sum_i f(X_i'\hat{\beta}) \cdot \hat{\beta}_j$. More representative but more computationally involved.
3. **Marginal effect at a specific X :** evaluate at a chosen profile (e.g. a typical observation in the policy-relevant subgroup).

The three numbers are usually similar but can differ when the distribution of $f(X_i'\hat{\beta})$ is highly skewed.

6.8 Asymptotic distribution of the MLE

Asymptotic normality via Taylor expansion

Let $S(\beta) = \partial \ln \mathcal{L} / \partial \beta$ and $H(\beta) = \partial^2 \ln \mathcal{L} / \partial \beta \partial \beta'$. The MLE solves $S(\hat{\beta}) = 0$. Taylor expand around the true β_0 :

$$0 = S(\hat{\beta}) \approx S(\beta_0) + H(\bar{\beta})(\hat{\beta} - \beta_0) \implies \hat{\beta} - \beta_0 \approx -[H(\bar{\beta})]^{-1} S(\beta_0).$$

Multiply by $\sqrt{n}$:

$$\sqrt{n}(\hat{\beta} - \beta_0) \approx -\left[\frac{1}{n}H(\bar{\beta})\right]^{-1} \cdot \frac{1}{\sqrt{n}}S(\beta_0).$$

By the LLN, $\frac{1}{n}H(\bar{\beta}) \rightarrow \mathbb{E}(H(\beta_0))$. By the CLT and $\mathbb{E}(S(\beta_0)) = 0$,

$$\frac{1}{\sqrt{n}}S(\beta_0) \xrightarrow{d} N(0, \mathbb{E}(S S')).$$

The **information matrix equality** $\mathbb{E}(H(\beta_0)) = -\mathbb{E}(S(\beta_0)S(\beta_0)')$ then collapses the sandwich into

$$\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, [-\mathbb{E}(H(\beta_0))]^{-1}) = N(0, [\mathbb{E}(S S')]^{-1}).$$

Problem Set Connection

PS4 Q2. The estimated variance for the discrete choice MLE is

$$\widehat{\text{Var}}(\hat{\beta}) = \left[\sum_i \frac{f_i^2}{F_i(1-F_i)} X_i X_i' \right]^{-1},$$

which is the inverse of the (estimated) information matrix — exactly what the asymptotic theory above gives. The derivation uses $\mathbb{E}[(y_i - F_i)^2] = F_i(1 - F_i)$ for the score outer product, plus the information matrix equality.

6.9 Goodness of fit and prediction

6.9.1 Pseudo R^2

In a linear regression, R^2 measures variance explained. With binary data this concept doesn't translate; instead, McFadden's pseudo- R^2 compares the fitted log-likelihood to that of a model

with only an intercept:

$$\text{pseudo-}R^2 = 1 - \frac{\ln \mathcal{L}(\hat{\beta})}{\ln \mathcal{L}(\bar{y})}.$$

A pseudo- R^2 between 0.2 and 0.4 is generally considered very good for cross-section discrete choice models.

6.9.2 Classification rule

To convert estimated probabilities $\hat{P}_i = F(X_i' \hat{\beta})$ into 0/1 predictions, you need a threshold. The naive choice is 0.5, but this performs poorly when the sample is imbalanced (e.g. 90% zeros). A better default is $\hat{y}_i = 1$ if $\hat{P}_i \geq \bar{y}$, where $\bar{y}$ is the sample proportion of ones. This matches the decision rule to the data's base rate.

6.10 Hypothesis testing

6.10.1 Likelihood Ratio test

- Estimate the model with restrictions imposed: log-likelihood $\hat{L}_R$.
- Estimate without restrictions: log-likelihood $\hat{L}_U$.
- Compute $\lambda = \hat{L}_R / \hat{L}_U$. Under H_0 , $-2 \ln \lambda$ asymptotically follows χ^2 with degrees of freedom equal to the number of restrictions.

6.10.2 Lagrange Multiplier test

When restricted estimation is easy but unrestricted is not, the LM test is appealing because it requires only the restricted estimate $\hat{\theta}_R$:

$$\text{LM} = \left[\frac{\partial \ln \mathcal{L}}{\partial \hat{\theta}_R} \right]' \cdot [-H(\hat{\theta}_R)]^{-1} \cdot \left[\frac{\partial \ln \mathcal{L}}{\partial \hat{\theta}_R} \right] \sim \chi^2_{(\# \text{restrictions})}.$$

The LM test asks: at the restricted estimate, how far is the score from zero? If far (in metric of the information matrix), the restrictions are likely false.

Problem Set Connection

PS4 Q6. Comparing LPM, logit, and probit estimates on female LFP data:

- All three give the same *signs* for coefficients — a useful sanity check.
- The probit coefficient on “children under 6” is 0.868 with positive sign. This is suspicious if you expected young children to reduce LFP. Possible explanations: omitted variable bias (e.g. rich families have more young children and the wife works), or the variable is mis-coded, or there really is a positive selection effect in the sample.
- For the probability calculation: $X' \hat{\beta} = -0.15 \cdot 20 + 0.131 \cdot 10 + 0.123 \cdot 10 - 0.019 \cdot 100 + 0.052 \cdot 30 + 0.868 \cdot 1 + 0.290$. Plug into $\Phi(\cdot)$. Whether it exceeds 0.5 depends on the sign of the index.
- In the LPM, the marginal effect of a young child is constant: each additional child under 6 raises predicted participation by 0.262, regardless of the family's other characteristics. In the probit, the same marginal effect depends on $\phi(X' \hat{\beta})$ and so varies with the family's profile.

Problem Set Connection

PS4 Q7. Contingent valuation. Each individual is asked “Are you willing to pay $\$w_i$?”. The latent willingness to pay is $v_i = X_i'\beta + \varepsilon_i$. They say yes iff $v_i \geq w_i$. Then

$$P(\text{yes}_i = 1 \mid X_i) = P(\varepsilon_i \geq w_i - X_i'\beta) = 1 - F_\varepsilon(w_i - X_i'\beta).$$

This is exactly a binary discrete choice model with the special feature that w_i varies experimentally across individuals — it acts as a known regressor with coefficient fixed at -1 . Estimating β gives the willingness-to-pay structure.

7. Multinomial Models

7.1 Setting

When Y_i takes one of $m > 2$ unordered categories, binary probit/logit no longer applies. The framework generalises: now we model

$$P_{ij} = P(Y_i = j) = P(Y_{ij} = 1), \quad j = 1, 2, \dots, m, \quad \sum_{j=1}^m P_{ij} = 1.$$

We use the dummy encoding $Y_{ij} = \mathbf{1}\{Y_i = j\}$. The probabilities are functions of regressors X and parameters θ : $P_{ij} = F_j(X; \theta)$.

7.2 Two types of regressors

The distinction matters more than it first appears.

- **Alternative-variant** X_{ij} : takes a different value for each alternative j within individual i . *Example:* the price of each heating fuel for a given household. The price of natural gas, the price of coal, and the price of electricity are three different numbers for the same household.
- **Alternative-invariant** X_i : same across alternatives for individual i . *Example:* household income or climate zone — one number per household, the same regardless of which heating choice we are evaluating.

These two types of variables enter the model in fundamentally different ways, which gives rise to three named special cases:

Regressors	Model	Probability formula
Alternative-variant only	Conditional logit	$P_{ij} = e^{X_{ij}'\beta} / \sum_{\ell} e^{X_{i\ell}'\beta}$
Alternative-invariant only	Multinomial logit	$P_{ij} = e^{X_i'\beta_j} / \sum_{\ell} e^{X_i'\beta_{\ell}}$
Both	Mixed logit	$P_{ij} = e^{W_i'\beta_j + X_{ij}'\alpha} / \sum_{\ell} e^{W_i'\beta_{\ell} + X_{i\ell}'\alpha}$

The key visible difference: in conditional logit, β is a single vector shared across alternatives, and identification comes from variation in X_{ij} across j . In multinomial logit, each alternative gets its own coefficient vector β_j , and identification comes from how the same X_i pushes households toward different choices.

7.3 Random utility foundation

Concept

McFadden's framework: the latent utility individual i derives from alternative j is $U_{ij} = \mu_{ij} + \varepsilon_{ij}$, with μ_{ij} depending on observed regressors and ε_{ij} unobserved. Individual i chooses alternative j iff $U_{ij} \geq U_{ik}$ for all $k \neq j$. The choice probability is

$$P_{ij} = P(U_{ij} \geq U_{ik} \forall k \neq j) = P(\varepsilon_{ik} - \varepsilon_{ij} \leq \mu_{ij} - \mu_{ik} \forall k \neq j).$$

Distributional assumption. If the ε_{ij} are i.i.d. **Type-I extreme value (Gumbel)** with CDF $F(\varepsilon) = e^{-e^{-\varepsilon}}$, then the choice probabilities take the elegant closed form

$$P_{ij} = \frac{e^{\mu_{ij}}}{\sum_{\ell} e^{\mu_{i\ell}}}.$$

This is the foundation of the logit family. McFadden's original application was to commute mode choice in the San Francisco Bay Area (BART study).

7.4 Conditional logit

For alternative-variant regressors only, $\mu_{ij} = X'_{ij}\beta$ and

$$P_{ij} = \frac{e^{X'_{ij}\beta}}{\sum_{\ell=1}^m e^{X'_{i\ell}\beta}}.$$

A single β vector applies across alternatives. The probabilities sum to 1 by construction:

$$\sum_{j=1}^m P_{ij} = \frac{\sum_j e^{X'_{ij}\beta}}{\sum_{\ell} e^{X'_{i\ell}\beta}} = 1.$$

7.4.1 Independence of Irrelevant Alternatives (IIA)

A famous property of any logit-family model:

$$\frac{P_{ij}}{P_{ik}} = \frac{e^{X'_{ij}\beta} / \sum_{\ell} e^{X'_{i\ell}\beta}}{e^{X'_{ik}\beta} / \sum_{\ell} e^{X'_{i\ell}\beta}} = \frac{e^{X'_{ij}\beta}}{e^{X'_{ik}\beta}}.$$

The ratio of two choice probabilities depends *only* on the characteristics of those two alternatives, not on any third option.

Common Pitfall

The red bus / blue bus paradox. Suppose you choose between car (probability 1/3) and red bus (probability 2/3), so $P(\text{red bus}) = 2P(\text{car})$. Now introduce a blue bus, which is identical to the red bus except for colour. Common sense says the new probabilities should be 1/3 each: car, red bus, blue bus.

But the IIA logit model preserves the 2:1 ratio between red bus and car *and* adds another factor-of-2 between blue bus and car. Working through the algebra, the model predicts $\{1/5, 2/5, 2/5\}$. This is unreasonable: introducing a near-perfect substitute for the red bus should not change car's market share at all.

When alternatives are close substitutes, IIA fails, and you should use nested logit, multinomial probit, or mixed logit.

7.4.2 Conditional logit FOC

Score function for conditional logit

The log-likelihood is $\ln \mathcal{L} = \sum_i \sum_j y_{ij} \ln P_{ij}$. We need $\partial P_{ij} / \partial \beta$:

$$\frac{\partial P_{ij}}{\partial \beta} = \frac{e^{X'_{ij}\beta} X_{ij} \sum_{\ell} e^{X'_{i\ell}\beta} - e^{X'_{ij}\beta} \sum_{\ell} e^{X'_{i\ell}\beta} X_{i\ell}}{(\sum_{\ell} e^{X'_{i\ell}\beta})^2} = P_{ij} \left(X_{ij} - \sum_{\ell} P_{i\ell} X_{i\ell} \right).$$

The FOC is

$$\frac{\partial \ln \mathcal{L}}{\partial \beta} = \sum_i \sum_j \frac{y_{ij}}{P_{ij}} \cdot P_{ij} \left(X_{ij} - \sum_{\ell} P_{i\ell} X_{i\ell} \right) = \sum_i \sum_j (y_{ij} X_{ij} - y_{ij} \sum_{\ell} P_{i\ell} X_{i\ell}).$$

Since $\sum_j y_{ij} = 1$ for each i , the second sum is $\sum_i \sum_{\ell} P_{i\ell} X_{i\ell}$. So the FOC simplifies beautifully:

$$\sum_i \sum_j (y_{ij} - P_{ij}) X_{ij} = 0.$$

Interpretation: the model fits sample averages of choice indicators to model-implied probabilities, weighted by alternative characteristics. This is a method-of-moments style condition.

7.5 Multinomial logit

When regressors are alternative-invariant, $\mu_{ij} = X'_i \beta_j$ and

$$P_{ij} = \frac{e^{X'_i \beta_j}}{\sum_{\ell} e^{X'_i \beta_{\ell}}}.$$

Each alternative now has its own coefficient vector β_j .

7.5.1 The identification problem**Why we must normalise one β_j to zero**

Suppose $(\beta_1^*, \dots, \beta_m^*)$ is an MLE. Consider an alternative parameterisation $\beta_j = \beta_j^* + c$ for all j , where c is some fixed vector. Then

$$P_{ij}|_{\beta_h+c} = \frac{e^{X'_i(\beta_j^*+c)}}{\sum_{\ell} e^{X'_i(\beta_{\ell}^*+c)}} = \frac{e^{X'_i c} \cdot e^{X'_i \beta_j^*}}{e^{X'_i c} \cdot \sum_{\ell} e^{X'_i \beta_{\ell}^*}} = P_{ij}|_{\beta_h^*}.$$

The constant c cancels. Adding any vector to all coefficients gives the same probabilities, hence the same likelihood — so the model is over-identified by exactly one β vector worth of parameters.

Solution. Set $\beta_1 = 0$ for one chosen “reference” alternative. Then

$$P_{i1} = \frac{1}{1 + \sum_{\ell=2}^m e^{X'_i \beta_{\ell}}}, \quad P_{ij} = \frac{e^{X'_i \beta_j}}{1 + \sum_{\ell=2}^m e^{X'_i \beta_{\ell}}} \text{ for } j \geq 2.$$

Equivalently, the log odds-ratios become readable:

$$\ln \frac{P_{ij}}{P_{i1}} = X'_i \beta_j.$$

Coefficient β_{jk} measures how a one-unit rise in X_k changes the log odds of choosing alternative j over alternative 1.

7.5.2 MNL marginal effects and FOC

Multinomial logit derivatives

Write $\delta_{jk} = \mathbf{1}\{j = k\}$. The two key results are:

$$\frac{\partial P_{ij}}{\partial X_{ik}} = P_{ij}(\delta_{jk} - P_{ik})\beta, \quad (2)$$

$$\frac{\partial \ln \mathcal{L}}{\partial \beta_k} = \sum_i (y_{ik} - P_{ik})X_i = 0, \quad (3)$$

$$\text{Var}(\hat{\beta}) = \left[\sum_i P_{ij}(\delta_{jk} - P_{ik})X_i X_i' \right]^{-1}. \quad (4)$$

The first follows from product/quotient rules on the exponential expression. The second comes from substituting the first into $\partial \ln \mathcal{L} / \partial \beta_k = \sum_i \sum_j (y_{ij} / P_{ij})(\partial P_{ij} / \partial \beta_k)$ and simplifying using $\sum_j y_{ij} = 1$. The third is the inverse of the information matrix.

Problem Set Connection

PS4 Q8 walkthrough.

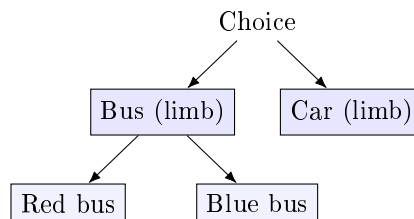
- (a) $\partial P_{ij} / \partial \beta_k$. Use the quotient rule on $P_{ij} = e^{X_i' \beta_j} / D$ where $D = \sum_\ell e^{X_i' \beta_\ell}$. Differentiating with respect to β_k :

$$\frac{\partial P_{ij}}{\partial \beta_k} = \frac{\delta_{jk} e^{X_i' \beta_j} X_i \cdot D - e^{X_i' \beta_j} \cdot e^{X_i' \beta_k} X_i}{D^2} = P_{ij}(\delta_{jk} - P_{ik})X_i.$$

- (b) $\partial \ln \mathcal{L} / \partial \beta_k$. Plug part (a) into the score formula and simplify. The P_{ij} 's cancel cleanly because of the structure of $\delta_{jk} - P_{ik}$, giving $\sum_i (y_{ik} - P_{ik})X_i$.
- (c) $\text{Cov}(\hat{\beta}_j, \hat{\beta}_k)$. The information matrix has block structure with diagonal block $\sum_i P_{ij}(1 - P_{ij})X_i X_i'$ and off-diagonal blocks $-\sum_i P_{ij}P_{ik}X_i X_i'$. Combine these as $\sum_i P_{ij}(\delta_{jk} - P_{ik})X_i X_i'$. Inverting gives the covariance matrix.

7.6 Nested logit

Nested logit relaxes IIA in a structured way: alternatives are organised into nests (limbs) and branches.



The errors ε_{jk} are correlated within each limb but independent across limbs. The choice probability decomposes as

$$P_{jk} = P_j \cdot P_{k|j},$$

where P_j is the probability of choosing limb j and $P_{k|j}$ is the conditional probability of branch k given limb j .

Probability formulas. Let Z_j be variables that vary across limbs and X_{jk} vary within branches.

Define the **inclusive value**

$$I_j = \ln \left(\sum_{h=1}^{K_j} e^{X'_{jh} \beta_j / \rho_j} \right),$$

which summarises the attractiveness of all options in limb j . Then

$$P_j = \frac{e^{Z'_j \alpha + \rho_j I_j}}{\sum_m e^{Z'_m \alpha + \rho_m I_m}}, \quad P_{k|j} = \frac{e^{X'_{jk} \beta_j / \rho_j}}{\sum_{\ell} e^{X'_{j\ell} \beta_j / \rho_j}}.$$

Intuition

The parameter $\rho_j \in (0, 1]$ controls within-limb correlation:

- $\rho_j = 1$: errors uncorrelated within the limb $\Rightarrow$ no nesting effect, the model collapses back to standard logit.
- $\rho_j \rightarrow 0$: errors highly correlated within the limb $\Rightarrow$ alternatives within the limb are near-perfect substitutes (e.g. red and blue bus).

This is exactly what was needed to fix the red/blue bus paradox: red and blue bus go in the same limb with $\rho \rightarrow 0$, so their joint share doesn't bleed into car's share.

7.7 Multinomial probit

If we replace the i.i.d. Gumbel errors with multivariate normal $\varepsilon \sim N(0, \Sigma)$, the choice probabilities no longer have a closed form but the IIA restriction disappears. The covariance matrix Σ becomes a parameter to estimate, allowing rich substitution patterns. The cost is computational: evaluating the likelihood requires high-dimensional normal integrals (typically via simulation methods such as the GHK algorithm).

7.8 Mixed logit (random coefficients)

A different relaxation of IIA: let coefficients vary across individuals.

$$U_{ij} = X'_{ij} \beta_i + \varepsilon_{ij}, \quad \beta_i \sim N(\beta, \Sigma_\beta).$$

Each individual has her own coefficient draw, and we observe only her choices — not her β_i . The implied choice probability for individual i is

$$P_{ij} = \int L_{ij}(\beta) f(\beta) d\beta, \quad L_{ij}(\beta) = \frac{e^{X'_{ij} \beta}}{\sum_{\ell} e^{X'_{i\ell} \beta}}.$$

Estimation proceeds by simulation: draw R values of β from $N(\hat{\beta}, \hat{\Sigma}_\beta)$, evaluate $L_{ij}(\beta^{(r)})$ for each, average. Maximise the simulated log-likelihood over β and Σ_β .

Mixed logit can approximate any random utility model and accommodates correlation patterns that IIA logit cannot. It is the modern workhorse in applied discrete-choice work.

8. Ordered Models

8.1 The setup

Some categorical outcomes carry natural ordering: “bad / OK / good”, survey responses on a 1–5 Likert scale, education attainment by level. Multinomial logit ignores this structure; ordered probit/logit exploits it.

Concept

Latent variable model. There is an unobserved continuous variable $y_i^* = X_i'\beta + \varepsilon_i$ representing, say, “true” health. We observe a discrete category y_i that depends on which interval y_i^* falls into:

$$y_i = \begin{cases} 0 & \text{if } y_i^* < 0, \\ 1 & \text{if } 0 < y_i^* < c_1, \\ 2 & \text{if } c_1 < y_i^* < c_2, \\ \vdots & \\ J & \text{if } c_{J-1} < y_i^*. \end{cases}$$

The cutoffs $c_1 < c_2 < \dots < c_{J-1}$ are unknown parameters to estimate alongside β .

8.2 Probabilities and likelihood

Assume $\varepsilon_i \sim N(0, 1)$ (ordered probit). The probabilities follow directly from the latent model:

$$\begin{aligned} P(y_i = 0) &= P(\varepsilon_i < -X_i'\beta) = \Phi(-X_i'\beta), \\ P(y_i = 1) &= \Phi(c_1 - X_i'\beta) - \Phi(-X_i'\beta), \\ &\vdots \\ P(y_i = J) &= 1 - \Phi(c_{J-1} - X_i'\beta). \end{aligned}$$

Each probability is the area under the standard normal curve between two cutoffs $c_{j-1} - X_i'\beta$ and $c_j - X_i'\beta$. Higher $X_i'\beta$ shifts the whole distribution to the right, raising the probability of higher categories.

The log-likelihood is

$$\ln \mathcal{L} = \sum_i \sum_j y_{ij} \ln P_{ij},$$

and maximisation gives $\hat{\beta}, \hat{c}_1, \dots, \hat{c}_{J-1}$.

Problem Set Connection**PS4 Q3.**

- (a) The probabilities are exactly as above with $J = 2$ and one interior cutoff b :

$$P(y_i = 0) = \Phi(-\alpha - \beta x_i), \quad P(y_i = 1) = \Phi(b - \alpha - \beta x_i) - \Phi(-\alpha - \beta x_i), \quad P(y_i = 2) = 1 - \Phi(b - \alpha - \beta x_i).$$

The likelihood is the product of these probabilities raised to indicator powers y_{ij} .

- (b) Suppose we used cutoffs $a < b$ instead of $0 < b$:

$$y_i = \begin{cases} 0 & y_i^* \leq a \\ 1 & a < y_i^* \leq b \\ 2 & b < y_i^* \end{cases}$$

Then $P(y_i = 0) = \Phi(a - \alpha - \beta x_i)$. Define $\tilde{\alpha} = \alpha - a$. Substituting,

$$\Phi(a - \alpha - \beta x_i) = \Phi(-\tilde{\alpha} - \beta x_i).$$

The probabilities depend only on $\tilde{\alpha} = \alpha - a$ — never on α and a separately. Hence a and α are not jointly identified. The standard fix is to set $a = 0$ (as in part (a)) so α is identified.

8.3 Applications

A few real-world ordered-choice studies mentioned in lecture:

- Thomas Miles (2005): does appearing on America's Most Wanted change criminal arrest outcomes (an ordered measure of severity)?
- Health behaviours of elderly individuals — Chinese Longitudinal Health Longevity Survey.
- Bresnahan and Reiss (1991) on the determinants of market entry: how many firms enter is an ordered count.

9. Censored and Truncated Models

9.1 The data problem

Concept

Censored data: for some observations, we see X_i but not the true value of Y_i — either because Y_i is recorded as a constant (e.g. income recorded as “\$250k+”), or because Y_i^* would have been negative but is reported as zero (e.g. hours worked).

Truncated data: for some observations, we see neither X_i nor Y_i — the entire row is missing. If observations are missing at random, this is harmless. If missingness depends on the variables of interest (e.g. only employed people respond to a survey on hours worked), it is a serious selection problem.

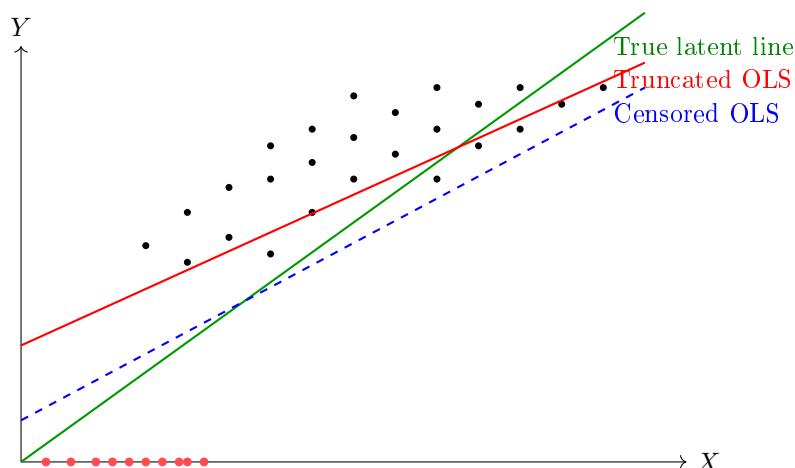
9.2 Examples

- **Hours worked.** If unemployed, $Y_i = 0$. The latent “desired hours” Y_i^* may be negative (they would actually pay to be at home), but we record only the non-negative observation. Censored.
- **Concert ticket sales.** The stadium has a capacity. If demand exceeds capacity, sales are recorded at capacity, not at true demand. Censored from above.
- **Income survey.** Above some threshold (\$250k say), the value is recorded as “\$250k+”. Censored from above.
- **Gated communities.** Wealthy residents may not respond to the survey at all. Truncated from above (selection on the dependent variable).
- **Household budget survey: healthcare spending.** A large share of households report zero spending on a particular category in any given month. The latent “willingness to spend” may be positive but small enough that no purchase happens. Censored at zero.

9.3 Why OLS fails: the latent-variable picture

Consider the latent model $y_i^* = X_i' \beta + u_i$. The observed variable is

$$y_i = \begin{cases} y_i^* & \text{if } y_i^* > 0, \\ 0 \text{ (censored) or unobserved (truncated)} & \text{otherwise.} \end{cases}$$

Diagram of the bias

The green line is the true latent regression $\mathbb{E}(y^*|X) = X'\beta$. The red line is what OLS estimates if you drop the zeros (truncated regression bias — too flat). The dashed blue line is what OLS estimates if you keep the zeros (censored regression bias — intermediate, still wrong).

9.4 The truncated regression model

If we keep only observations with $y_i^* > 0$, the conditional expectation in the truncated sample is

$\mathbb{E}(Y_i | Y_i \text{ observed})$ in the truncated model

$$\begin{aligned}\mathbb{E}(Y_i | X_i, Y_i \text{ observed}) &= \mathbb{E}(y_i^* | y_i^* > 0, X_i) \\ &= \mathbb{E}(X_i'\beta + u_i | X_i'\beta + u_i > 0, X_i) \\ &= X_i'\beta + \mathbb{E}(u_i | u_i > -X_i'\beta).\end{aligned}$$

Assuming $u_i \sim N(0, \sigma^2)$, the truncated conditional mean of u_i/σ is the well-known inverse Mills ratio:

$$\mathbb{E}(u_i | u_i > -X_i'\beta) = \sigma \cdot \frac{\phi(X_i'\beta/\sigma)}{\Phi(X_i'\beta/\sigma)} = \sigma\lambda_i.$$

So

$$\mathbb{E}(Y_i | X_i, Y_i \text{ observed}) = X_i'\beta + \sigma\lambda_i \neq X_i'\beta.$$

The OLS regression on truncated data has an omitted variable: the inverse Mills ratio. Since λ_i depends on X_i , this omitted variable is correlated with the included regressors, biasing $\hat{\beta}$.

9.4.1 The inverse Mills ratio

$$\lambda_i = \frac{\phi(X_i'\beta/\sigma)}{\Phi(X_i'\beta/\sigma)}.$$

Intuition

λ_i is the conditional mean of a standard normal random variable, given that it exceeds the threshold $-X_i'\beta/\sigma$. Properties:

- $\lambda_i > 0$ always (the truncated mean of a standard normal must be positive, given the truncation point).

- As $X_i'\beta/\sigma \rightarrow \infty$ (truncation barely binds), $\lambda_i \rightarrow 0$.
- As $X_i'\beta/\sigma \rightarrow -\infty$ (almost everyone is truncated out), λ_i grows large.

9.4.2 MLE for the truncated model

The density of y_i given that it's observed is the truncated normal density:

$$f(y_i | y_i^* > 0, X_i) = \frac{(1/\sigma) \phi((y_i - X_i'\beta)/\sigma)}{\Phi(X_i'\beta/\sigma)}.$$

The denominator rescales the density to integrate to one over the truncated range. The log-likelihood is

$$\ln \mathcal{L} = \sum_i \left[\ln \frac{1}{\sigma} + \ln \phi\left(\frac{y_i - X_i'\beta}{\sigma}\right) - \ln \Phi\left(\frac{X_i'\beta}{\sigma}\right) \right].$$

Maximise over β and σ to get $\hat{\beta}_{\text{MLE}}, \hat{\sigma}_{\text{MLE}}$.

9.4.3 Marginal effect (truncated model)

$$\partial \mathbb{E}(Y_i | X_i, Y_i \text{ observed}) / \partial X_{ij}$$

Differentiating $\mathbb{E}(Y_i | X_i, \text{obs}) = X_i'\beta + \sigma\lambda_i$:

$$\frac{\partial \mathbb{E}(Y_i | \text{obs})}{\partial X_{ij}} = \beta_j + \sigma \frac{\partial \lambda_i}{\partial X_{ij}}.$$

Using $\partial \lambda / \partial z = -z\lambda - \lambda^2$ where $z = X_i'\beta/\sigma$:

$$\frac{\partial \lambda_i}{\partial X_{ij}} = - \left(\frac{X_i'\beta}{\sigma} \lambda_i + \lambda_i^2 \right) \cdot \frac{\beta_j}{\sigma}.$$

Substituting:

$$\frac{\partial \mathbb{E}(Y_i | \text{obs})}{\partial X_{ij}} = \left(1 - \frac{X_i'\beta}{\sigma} \lambda_i - \lambda_i^2 \right) \beta_j.$$

The factor in parentheses is always between 0 and 1 — the truncation *shrinks* the marginal effect compared to the latent model.

9.5 The censored regression model

In the censored case we keep all observations but $y_i = 0$ when $y_i^* \leq 0$. The unconditional expectation:

$$\mathbb{E}(Y_i | X_i) \text{ in the censored model}$$

$$\begin{aligned} \mathbb{E}(Y_i | X_i) &= \mathbb{E}(Y_i | y_i^* \leq 0) \cdot \text{P}(y_i^* \leq 0) + \mathbb{E}(Y_i | y_i^* > 0) \cdot \text{P}(y_i^* > 0) \\ &= 0 + (X_i'\beta + \sigma\lambda_i) \cdot \Phi(X_i'\beta/\sigma) \\ &= X_i'\beta \cdot \Phi(X_i'\beta/\sigma) + \sigma\phi(X_i'\beta/\sigma), \end{aligned}$$

where we used $\sigma\lambda_i \cdot \Phi(\cdot) = \sigma\phi(\cdot)$. This is *not* equal to $X_i'\beta$. OLS run on all observations (including zeros) is biased.

9.6 The Tobit model

The **Tobit** (Tobin's probit) model combines the censoring rule and the regression equation into a single likelihood. The probability of being at the corner is computed using probit; the probability density above the corner is computed using truncated regression.

Tobit log-likelihood

For each observation, contribute either the censoring probability or the density:

$$\begin{aligned} P(y_i = 0 \mid X_i) &= P(y_i^* \leq 0) = 1 - \Phi(X_i' \beta / \sigma), \\ f(y_i \mid X_i) &= \frac{1}{\sigma} \phi((y_i - X_i' \beta) / \sigma) \quad \text{for } y_i > 0. \end{aligned}$$

The log-likelihood is

$$\ln \mathcal{L} = \sum_i \mathbf{1}\{y_i = 0\} \ln(1 - \Phi(X_i' \beta / \sigma)) + \mathbf{1}\{y_i \neq 0\} \ln\left(\frac{1}{\sigma} \phi((y_i - X_i' \beta) / \sigma)\right).$$

Reparametrising with $\gamma = \beta / \sigma$ and $\theta = 1 / \sigma$ makes the likelihood linear in the parameters and easier to optimise.

9.6.1 Two marginal effects

The Tobit model gives two marginal effects, depending on which expectation we differentiate:

- **Conditional marginal effect (on the truncated/observed sample):**

$$\frac{\partial \mathbb{E}(Y_i \mid Y_i > 0, X)}{\partial X_{ij}} = \left(1 - \frac{X_i' \beta}{\sigma} \lambda_i - \lambda_i^2\right) \beta_j.$$

Says: among those for whom $Y_i > 0$, how does X shift the level of Y ?

- **Unconditional marginal effect (on the whole sample):**

$$\frac{\partial \mathbb{E}(Y_i \mid X)}{\partial X_{ij}} = \beta_j \cdot \Phi(X_i' \beta / \sigma).$$

Says: how does X shift the unconditional mean of Y , accounting for both the chance of leaving the corner and the level effect?

The unconditional ME is generally smaller in magnitude because some of the predicted mass is at zero.

9.7 Heckman's two-step (sample selection)

9.7.1 When the censoring mechanism differs from the outcome

The Tobit assumes the latent variable that determines whether $y_i = 0$ is the *same* as the latent variable whose value we observe when $y_i > 0$. This is restrictive. In many settings, two distinct mechanisms are at work:

- One process determines whether Y is observed at all (selection).
- A separate process determines the value of Y given that it's observed (outcome).

Female labour earnings. Whether a woman participates in the labour market depends on factors like reservation wage, child care, household bargaining — captured by some variables Z_i . If she works, her earnings depend on education, experience, etc. — captured by X_i . The two equations have potentially different regressors.

Concept**Two-equation Heckman model:**

$$\text{Outcome: } y_i^* = X_i' \beta + u_i,$$

$$\text{Selection: } d_i = \mathbf{1}\{Z_i' \alpha + \varepsilon_i > 0\},$$

$$y_i = y_i^* \text{ if } d_i = 1, \text{ unobserved if } d_i = 0.$$

The errors (u_i, ε_i) are bivariate normal with $\text{Corr}(u_i, \varepsilon_i) = \rho$. Set $\sigma_\varepsilon = 1$ for identification (we cannot separately estimate σ_u and σ_ε , only their product).

9.7.2 The selection bias formula**Conditional mean in the selected sample**

For an observed individual ($d_i = 1$):

$$\begin{aligned} \mathbb{E}(Y_i | X_i, Z_i, \text{obs}) &= \mathbb{E}(y_i^* | \varepsilon_i > -Z_i' \alpha, X_i) \\ &= X_i' \beta + \mathbb{E}(u_i | \varepsilon_i > -Z_i' \alpha). \end{aligned}$$

If u_i and ε_i are independent ($\rho = 0$), the second term vanishes and OLS on the observed sample is unbiased. *This is rarely realistic.*

If $\rho \neq 0$, conditioning on the selection event $\varepsilon_i > -Z_i' \alpha$ creates non-zero expectation for the correlated u_i . Specifically, with bivariate normality:

$$\mathbb{E}(u_i | \varepsilon_i > -Z_i' \alpha) = \rho \sigma_u \cdot \frac{\phi(Z_i' \alpha)}{\Phi(Z_i' \alpha)} = \rho \sigma_u \cdot \lambda(Z_i' \alpha).$$

So

$$\mathbb{E}(Y_i | X_i, Z_i, \text{obs}) = X_i' \beta + \gamma \lambda_i, \quad \gamma = \rho \sigma_u.$$

OLS that ignores λ_i has selection bias proportional to ρ .

9.7.3 Heckman's two-step estimator

1. **Step 1.** Fit a probit on the full sample with selection indicator d_i and regressors Z_i :

$$d_i = \mathbf{1}\{Z_i' \alpha + \varepsilon_i > 0\}, \quad \varepsilon_i \sim N(0, 1).$$

Obtain $\hat{\alpha}$ and compute the inverse Mills ratio for each individual:

$$\hat{\lambda}_i = \frac{\phi(Z_i' \hat{\alpha})}{\Phi(Z_i' \hat{\alpha})}.$$

2. **Step 2.** Restrict to the observed sample ($d_i = 1$) and run OLS of Y_i on X_i and $\hat{\lambda}_i$:

$$Y_i = X_i' \beta + \gamma \hat{\lambda}_i + \text{error}.$$

Estimating γ corrects for sample selection. A test of $\gamma = 0$ is a test of no selection bias.

Problem Set Connection

PS4 Q5. Earnings of men with disabilities, with $u_i \sim \text{Uniform}[-1, 1]$ and the truncation rule $y_i^* > B$:

- **(a) Likelihood for the positive-earnings sample.** For an observation with $y_i^* > B$, the truncated density is

$$f(y_i | y_i^* > B) = \frac{f_u(y_i - X_i'\beta)}{P(y_i^* > B)} = \frac{1/2}{(1 - (B - X_i'\beta))/2} \cdot \mathbf{1}\{|y_i - X_i'\beta| < 1\} = \frac{1}{1 - B + X_i'\beta}.$$

Likelihood is the product over the truncated sample.

- **(b) Full-sample likelihood.** Each observation contributes either the censoring probability $P(y_i^* \leq B) = (1 - B + X_i'\beta)/2$ if at the corner, or the density above the corner. Combine with indicators just like the Tobit construction.

9.7.4 Two paths to identification

The Heckman model is technically identified even with $Z_i = X_i$, but only through the non-linearity of the inverse Mills ratio. This is fragile.

Common Pitfall

$\lambda(Z'\alpha)$ is non-linear, but over the bulk of its range it is approximately linear in $Z'\alpha$. If $Z_i = X_i$, then $\hat{\lambda}_i$ is nearly a linear function of X_i , so it is highly collinear with X_i in the second-stage regression. The result is unstable, near-singular estimates with huge standard errors.

In practice, you need an exclusion restriction: at least one variable in Z_i that is *not* in X_i . Such a variable affects the probability of selection but not the outcome conditional on selection. Common examples: number of children for women's labour supply, distance to school for educational attainment.

10. Panel Data

10.1 What panel data is and why it helps

Concept

Panel (longitudinal) data tracks the same entities — individuals, firms, countries — over multiple time periods. The double subscript (i, t) indexes entity i at time t , with $i = 1, \dots, n$ and $t = 1, \dots, T$, giving nT observations.

The killer feature: observing the same individual repeatedly lets us *difference out* time-invariant unobserved heterogeneity. With cross-section data, an unobserved trait correlated with the regressors creates omitted variable bias. With panel data, we can sweep that trait away.

10.1.1 Examples

- **SILC** (Survey of Income and Living Conditions): rotating panel where each year, one quarter of the sample drops out and is replaced. Each respondent stays four years.
- **Household Budget Survey:** typically *not* a panel — each year a fresh cross-section. Cannot follow households over time.
- **PSID, BHPS, Understanding Society:** long-running household panels.

10.1.2 Balanced vs unbalanced

- **Balanced:** all n entities observed in all T periods. Throughout this section we assume balanced.
- **Unbalanced:** entities observed for different numbers of periods. Estimation methods generalise but notation gets messier.

10.2 The error structure

The standard panel-data regression is

$$Y_{it} = X'_{it}\beta + u_{it}, \quad u_{it} = \alpha_i + \lambda_t + \varepsilon_{it}.$$

The composite error has three components:

- α_i : **individual-specific, time-invariant** effect. Captures all unobserved characteristics of entity i that don't change over the sample period (innate ability, firm management quality, country institutions).
- λ_t : **time-specific, entity-invariant** effect. Captures aggregate shocks that hit everyone at time t (recessions, policy changes).
- ε_{it} : idiosyncratic shock specific to (i, t) .

In matrix notation: $Y_i = X_i\beta + u_i$ where Y_i is $T \times 1$ and X_i is $T \times k$. Stacking across individuals, $Y = X\beta + u$ is $nT \times 1$.

10.3 Pooled OLS

The simplest approach: ignore the panel structure entirely. Treat all nT observations as a long cross-section and run OLS:

When is it OK?

- If $\mathbb{E}(u | X) = 0$: pooled OLS is consistent.
- If additionally $\mathbb{E}(uu' | X) = \sigma^2 I$: pooled OLS is efficient.

When does it fail?

1. **Inefficiency.** If α_i is present, errors for the same individual at different times are correlated: $\mathbb{E}(u_{it}u_{is}) = \sigma_\alpha^2 \neq 0$. Pooled OLS standard errors are then wrong; cluster-robust standard errors fix the inference but the estimator remains inefficient.
2. **Inconsistency.** If α_i is correlated with X_{it} (e.g. a person's innate ability correlates with their education), then $\mathbb{E}(u_{it} | X_{it}) = \mathbb{E}(\alpha_i | X_{it}) \neq 0$. The estimator is biased and inconsistent.

The fixed effects model handles the second, more serious problem.

10.4 Fixed effects model

Concept

The **fixed effects** model writes

$$Y_{it} = \alpha_i + X'_{it}\beta + \varepsilon_{it},$$

treating each α_i as an unknown parameter to be estimated (or eliminated). The crucial

feature: α_i is allowed to be *correlated* with X_{it} .

There are three equivalent ways to estimate β . They give the same answer; they differ in computational and conceptual convenience.

10.4.1 Approach 1: LSDV (Least Squares Dummy Variables)

Define a dummy for each individual: $d_{it}^j = \mathbf{1}\{i = j\}$. Then

$$Y_{it} = \alpha_1 d_{it}^1 + \alpha_2 d_{it}^2 + \cdots + \alpha_n d_{it}^n + X'_{it}\beta + \varepsilon_{it}.$$

This is just OLS with n extra dummy variables. Estimate by ordinary OLS, get $\hat{\alpha}_i$ and $\hat{\beta}$ at once.

Drawback: when n is large and T is small (the typical micro-panel), the $\hat{\alpha}_i$'s are not consistently estimated — there is only T data points per α_i . This is the **incidental parameters problem**. Importantly, $\hat{\beta}$ is still consistent in this case, even though the $\hat{\alpha}_i$'s are not.

10.4.2 Approach 2: Within (demeaning) estimator

The within transformation eliminates α_i

Take time-averages within each individual:

$$\bar{Y}_i = \frac{1}{T} \sum_{t=1}^T Y_{it} = \alpha_i + \bar{X}'_i \beta + \bar{\varepsilon}_i.$$

Subtract from the original equation:

$$Y_{it} - \bar{Y}_i = (X_{it} - \bar{X}_i)' \beta + (\varepsilon_{it} - \bar{\varepsilon}_i).$$

The α_i is gone — it appears in both lines and cancels. Now run OLS on the demeaned variables:

$$\hat{\beta}_{FE} = \arg \min_{\beta} \sum_{i,t} [(Y_{it} - \bar{Y}_i) - (X_{it} - \bar{X}_i)' \beta]^2.$$

10.4.3 Approach 3: First differences

Differencing eliminates α_i another way

Take first differences:

$$\Delta Y_{it} = Y_{it} - Y_{i,t-1} = (\alpha_i + X'_{it}\beta + \varepsilon_{it}) - (\alpha_i + X'_{i,t-1}\beta + \varepsilon_{i,t-1}) = \Delta X'_{it}\beta + \Delta \varepsilon_{it}.$$

The α_i cancels. Run OLS of ΔY_{it} on ΔX_{it} over $n \times (T - 1)$ observations.

Common Pitfall

First-differencing eliminates not just α_i but *all* time-invariant regressors. If your X includes variables that don't change over time (gender, country of birth), their coefficients cannot be estimated from a first-differenced model. Within (demeaning) has the same problem: a time-invariant X_{ij} has $X_{ij} - \bar{X}_{ij} = 0$ for all t .

This is a feature, not a bug: it's what allows fixed effects to control for unobserved heterogeneity. But it means you trade off the ability to estimate effects of time-invariant covariates.

Note on serial correlation in first differences. Even if the original ε_{it} are i.i.d., the differenced errors $\Delta\varepsilon_{it} = \varepsilon_{it} - \varepsilon_{i,t-1}$ have $\text{Cov}(\Delta\varepsilon_{it}, \Delta\varepsilon_{i,t-1}) = -\sigma_\varepsilon^2 \neq 0$ by construction. This affects standard errors; cluster-robust or Newey-West-style corrections are standard.

10.4.4 Within and first differences give the same $\hat{\beta}$

For $T = 2$, within and first differences are algebraically identical. For $T > 2$, they differ in finite samples but converge to the same probability limit. Within is more efficient under classical assumptions; first differences are more robust to certain forms of dynamic misspecification.

10.5 Random effects model

Concept

The **random effects** model also writes $Y_{it} = \alpha_i + X'_{it}\beta + \varepsilon_{it}$, but now treats α_i as a *random* component drawn from a distribution: $\alpha_i \sim N(\alpha, \sigma_\alpha^2)$, independent of X_{it} .

The key assumption is the **orthogonality** of α_i and X_{it} . If this holds, the random effects estimator is consistent *and* more efficient than fixed effects (it uses both the within and the between variation in the data). If it fails, RE is inconsistent while FE remains consistent.

10.5.1 The Hausman test

The standard tool for choosing between FE and RE is the Hausman test (not derived in detail in the lecture, but worth knowing):

- Under H_0 : $\alpha_i \perp X_{it}$. Both FE and RE are consistent, but RE is efficient. The two estimators should be close.
- Under H_A : α_i correlated with X_{it} . Only FE is consistent. The two estimators diverge.

The test statistic is

$$H = (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}})' [\widehat{\text{Var}}(\hat{\beta}_{\text{FE}}) - \widehat{\text{Var}}(\hat{\beta}_{\text{RE}})]^{-1} (\hat{\beta}_{\text{FE}} - \hat{\beta}_{\text{RE}}) \sim \chi_k^2.$$

Reject H_0 in favour of fixed effects if H is large.

Intuition

Practical decision tree:

- Suspect α_i correlates with regressors (e.g. unobserved ability with schooling)? Use FE.
- Comfortable assuming orthogonality and want efficiency? Use RE.
- Hausman test rejects orthogonality? Use FE.
- Need to estimate effects of time-invariant variables? RE (or hybrid models like Mundlak's correlated random effects).

10.6 Connection to the Mehic (2024) replication

This section deserves an explicit cross-link given that you are actively replicating Mehic's peer effects paper. The first-differencing transformation does two things at once:

1. Eliminates α_i — the unobserved individual fixed effect (e.g. “baseline academic ability” that doesn't change over the study period).

2. Eliminates *any* time-invariant regressor. So if you include controls like gender, family background, or initial cohort assignment, those coefficients become unidentified after differencing.

This is precisely the identification issue you flagged in our earlier conversation. The literature on dynamic panels (Arellano–Bond, system GMM) builds on the same first-differencing logic but uses lagged levels as instruments to handle the additional complication that lagged dependent variables become correlated with the differenced errors.

10.7 Closing summary of the panel-data toolkit

Summary

A compact decision summary:

- **Pooled OLS**: when error has no individual-specific component (rare in micro-data).
- **Random effects**: when α_i exists but is uncorrelated with X_{it} . More efficient than FE under this assumption.
- **Fixed effects** (within or first differences): when α_i is correlated with X_{it} . The conservative default.
- **Hausman test**: tells you which to use.
- **LSDV**: an equivalent way to estimate FE that also recovers individual effects (but inconsistent for α_i with small T).

For dynamic models with $Y_{i,t-1}$ on the right-hand side, fixed-effects estimators are biased when T is small (Nickell bias). The Arellano–Bond GMM family handles this and is the natural next topic — relevant to your econometrics replication work.

Final exam preparation: a quick checklist

A short list of derivations and concepts worth being able to reproduce on demand:

1. **ECM derivation from ADL(1,1)**. Subtract Y_{t-1} , add and subtract $\beta_0 X_{t-1}$. End up with $\Delta Y_t = \beta_0 \Delta X_t + (\rho_1 - 1)[Y_{t-1} - \phi X_{t-1}] + e_t$.
2. **Long-run multiplier formula**. For ADL(p, q): $(\beta_0 + \dots + \beta_q)/(1 - \rho_1 - \dots - \rho_p)$.
3. **Spurious regression**. Two independent random walks $\Rightarrow \text{Var}(u_t) \rightarrow \infty$, $\text{Corr}(u_t, u_{t+1}) \rightarrow 1$, OLS t -stats explode.
4. **Engle–Granger 2-stage estimator**. Stage 1 in levels, save residuals; Stage 2 in differences with lagged residual. Super-consistent in Stage 1, inference issues.
5. **VECM derivation from VAR(2)**. Subtract Y_{t-1} , add/subtract $(A_1 - I)Y_{t-2}$. Get $\Delta Y_t = m + B_1 \Delta Y_{t-1} - \Pi Y_{t-2} + e_t$ with $\Pi = I - A_1 - A_2$.
6. **Rank of Π = number of cointegrating relations**. Decomposition $\Pi = \alpha\beta'$.
7. **Johansen sequential test**. Trace and max statistics; sequential testing of $r = 0, 1, 2, \dots$
8. **Cholesky decomposition for IRFs**. $\Sigma_e = PP'$; $u = P^{-1}e$.
9. **LPM heteroskedasticity formula**. $\text{Var}(u_i) = p_i(1 - p_i)$.
10. **Probit/logit log-likelihood and FOC**. $\sum_i [(y_i - F_i)/(F_i(1 - F_i))] f_i X_i = 0$. Logit's simplification: $\sum_i (y_i - F_i) X_i = 0$.

11. **Marginal effect formula.** $f(X'\beta)\beta_j$ at the chosen X .
12. **Asymptotic variance.** $\widehat{\text{Var}}(\hat{\beta}) = [\sum_i f_i^2 / (F_i(1 - F_i)) X_i X_i']^{-1}$.
13. **Multinomial logit derivatives.** $\partial P_{ij} / \partial \beta_k = P_{ij}(\delta_{jk} - P_{ik})X_i$. FOC: $\sum (y_{ik} - P_{ik})X_i = 0$.
14. **IIA and the red-bus / blue-bus paradox.** Why nested logit, multinomial probit, or mixed logit are needed when alternatives are close substitutes.
15. **Inverse Mills ratio.** $\lambda(z) = \phi(z)/\Phi(z)$. Truncated mean: $\mathbb{E}(u \mid u > -X'\beta) = \sigma\lambda(X'\beta/\sigma)$.
16. **Tobit log-likelihood** as probit-plus-truncated-regression.
17. **Heckman 2-step.** Probit on selection, plug $\hat{\lambda}$ into outcome regression. Need exclusion restriction in practice.
18. **Within and first-difference fixed effects.** Both eliminate α_i . Cannot estimate effects of time-invariant regressors.
19. **Hausman test.** $\hat{\beta}_{\text{FE}}$ vs $\hat{\beta}_{\text{RE}}$ comparison; reject orthogonality if they differ.

End of guide. Good luck on the exam.